DOMAIN 2 - PROGRAM MANAGEMENT (QUESTIONS & ANSWERS)

Q249 A simple definition for hazard, as it applies to system safety, is:

1. A source of danger
2. Risk, peril, or jeopardy
3. A potential to do harm
4. A dangerous condition which can interfere with the orderly progress of an activity

Ans(3). potential to do harm is the simple and abbreviated version of the definition of "hazard." More formally, a hazard is "a potential condition, or set of conditions, either internal and/or external to a system, which when activated by a stimulus (set of events or conditions that transforms a hazard from its potential state to one that causes harm to the system) or series of events which culminate in a loss (accident)."

Q250 Of the following, which is the LEAST effective way for a manager to maximize his or her safety program?

1. Written safety policies
2. Supervisory safety training
3. Audit of safety program
4. Weekly staff meetings

Ans(4). Common elements to maximize a safety program include: written policies, supervisor training, auditing, investigation of all hazards and accidents, and formation of safety committees. Staff meetings can be on any schedule that allows you to reach your goals.

Q251 A procedure to make a job safer by identifying hazards in each step of the job and developing measures to counteract those hazards is a:

1. Time and motion analysis
2. Fault tree analysis
3. Job safety analysis
4. Probabilistic risk assessment

Ans (3). A job safety analysis examines job hazards during each step of the job.

Q252 What is a JSA?

1. A procedure of reviewing methods, hazards, and safety
2. A type of job description
3. A private company well-recognized by safety practitioners
4. Joint safety authority

Ans (1). A job safety analysis is a written procedure for reviewing job methods, uncovering hazards, and recommending safe procedures.

Q253 Which of these is an essential step in performing a JSA?

1. Break the job down into discreet steps
2. Formally identify the hazards in each discreet step
3. Compare the job to others
4. Both 1 and 2

Ans (4). The actual steps in a JSA are to identify the job, break the job down into steps, identify hazards, and determine how to control the hazards.

Q254 Military contractors for weapons research played a major role in developing:

1. Event tree and fault tree analysis
2. Environmental protection programs
3. Air sampling techniques
4. Noise measurement techniques

Ans (1). Event tree and fault tree analyses were developed for weapons systems in the 1950's and 1960's.

Q255 The industry where system safety techniques first were widely used was the:

1. Aerospace industry
2. Computer industry
3. Hazardous waste disposal industry
4. Steel industry

Ans(1).The aerospace industry developed the basic format of system safety. The government contracts issued to these companies required that systems safety be addressed in a quantitative manner. Military specifications have been written concerning system safety.

Q256 The industry where system safety techniques first were widely used was the:

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Ans(1).The aerospace industry developed the basic format of system safety. The government contracts issued to these companies required that systems safety be addressed in a quantitative manner. Military specifications have been written concerning system safety.

Q257 Risk-based inspection of pressure vessels involves:

1. Performing specific non-destructive examinations (NDEs) on specific intervals to understand the pressure vessel's current condition and to predict how the pressure vessel will degrade over the next inspection interval
2. Performing the specific ASME-required pressure vessel tests
3. Performing the specific OSHA-require pressure vessel tests
4. Determining whether a high-risk inspection is require based on the age of the pressure vessel and the vessel's service

Ans(1). Risk-based inspection (RBI) is a strategy gaining ground that considers the holistic effect of pressure vessel service against prescriptive requirements for pressure vessel safety. Sometimes the results of an RBI program recommend more frequent inspection intervals than are required by ASME, and sometimes the results of the RBI program recommend less frequent intervals. In many cases, an RBI program recommends tests in addition to those required by codes and regulations.

Q258 In the context of reliability-centered maintenance (RCM), what is a task selection guide?

1. A guide developed to help choose appropriate maintenance tasks based on the service of the equipment
2. A guide that describes the intervals between maintenance tasks
3. A guide that describes the actual maintenance tasks to perform on equipment
4. All of the above

Ans(4).RCM is a method to define an appropriate maintenance strategy based on the criticality of the equipment and the effectiveness of maintenance tasks to produce acceptable reliability performance. A task selection guide is developed to define the actual maintenance tasks and their intervals based on the type of service of the equipment. This information helps the maintenance and operations professionals to choose maintenance tasks at appropriate intervals.

Q259 Why would the following types of analyses be routinely conducted?
A) Vibration analysis
B) Lubricant analysis
C) Thermographic analysis

1. To determine how rotating equipment failed
2. To determine whether additional maintenance is needed on rotating equipment
3. To predict the day when rotating equipment will fail
4. To evaluate the overhaul of rotating equipment

Ans(2). Though any of these could be true, these types of condition monitoring tasks are typically associated with reliability-centered maintenance and help maintenance professionals determine when more extensive maintenance is needed on rotating equipment.

Q260 What is meant when a system is said to be "two fault tolerant?"

1. The system will fail when it sustains two independent failures.
2. The system will fail when it sustains three independent failures.
3. The system is designed to tolerate two independent failures and still operate.
4. The system is designed to tolerate three independent failures and still operate.

Ans(3). Fault-tolerance is a method of designing equipment to sustain one or more independent failures and still operate (or be able to transition into a fail-safe mode). In this case, the system is designed to sustain two independent failures and still function. Fault tolerance has nothing to do with when the system will fail (i.e., the system may sustain more than three independent failures and still function).

Q261 The primary reason for an accident investigation is to:

1. Find and reprimand those responsible
2. Provide OSHA with the accident facts
3. Prepare insurance forms to recover losses
4. Correct the conditions that caused the accident

Ans(4). The main purpose of an accident investigation is to identify the reason for the accident and correct it.

Q262 A Gantt chart is used in manufacturing to:

1. Show product development
2. Schedule work
3. Indicate the competitor's product quality
4. Display management structure

Ans(2). Gantt charts show how long each production process should take and when it should take place. It is widely used in project scheduling.

Q263 The most appropriate way to display critical quantitative data necessary for an operator to make a decision would be to use a:

1. Moving pointer
2. Fixed pointer
3. Printer
4. Flashing light

Ans(3). Of these four choices, the most appropriate way to display critical quantitative data would be to use a printer. There are better ways than a printer; however, of the choices offered, printing critical quantitative data is the best.

Q264 Laboratory quality control samples control which type of error?

1. Random
2. Systematic
3. Sampler
4. Preservation

Ans(2). These quality control samples include spiked samples and blanks. This keeps the instrument in calibration during the analytical series. Equipment errors are systematic errors.

Q265 Evaluation of safety and health program effectiveness is based on:

1. Frequency and severity rate comparisons
2. Quantitative criteria measures
3. Workers' compensation rates
4. ANSI Z16.4 - 1977

Ans(2). Sometimes, attempts to control accidents and their consequences can best be described as trial and error, chiefly because adequate measures of the effectiveness of controls do not exist in practice. Controls begin with quantifiable criteria measures. The degree to which accident control (the underlying purpose of a safety and health program) is possible is a function of the adequacy of the measures used to identify the type and magnitude of potential injury-producing situations. To do this effectively, a variety of quantitative measures are needed.

Q266 Of the following, what is the greatest benefit to using a computer to process historical, safety-related data (accidents, injuries, illnesses, etc.)?

1. The computer can quickly perform complicated math and generate useful trend data and analyses
2. The computer can easily generate reports of high quality and consistency for management use
3. The computer can maintain the names and locations of supervisors for efficient incident report notification
4. The computer can distribute reports electronically for faster dissemination to employees

Ans(1). Though computers clearly assist in all of these areas, its ability to do complicated math for trend analyses allows the safety practitioner to rapidly identify trends and address these trends with the relevant supervisors and managers.

Q267 In any study involving a statistical analysis, the main focus is on sets of numerical data. The entire set of data from which your data came is called the:

1. Sample
2. Population
3. Bell shaped curve
4. Normal distribution

Ans(2). The sample is what you take. You hope (and assume) that your sample is representative of the population from which it was drawn.

Q268 Air sampling data are log normally distributed. To analyze air sampling data using a t-test one must:

1. Use the log of the concentration
2. Convert to an exponential distribution
3. Use a Mantel-Haenzel test instead
4. Make no changes to use the t-test

Ans(1). A t-test is used with a normal distribution. To analyze log normally distributed data, one must convert the air sampling data to logarithms to use the t-test. Converting to logarithms should result in a normal distribution.

Q269 Which of the following components are not specifically required in a process hazard analysis?

A. Facility inspection procedures
B. Facility siting
C. Human factors
D. The hazards of the process
E. An assessment of the worst-case release into the environment of the covered process materials
F. The identity of engineering and administrative controls for specific process hazards
G. Consequences of failures of engineering controls
H. Consequences of failures of administrative controls

1. E, F, G, H
2. A, F, G, H
3. A, E
4. B, C

Ans(3). These items are not specifically required to be assessed in a process hazard analysis; however, depending upon the actual analysis, these items may come up.

Q270 When you are interviewing a person associated with a recent high-consequence incident, what fundamental law of human behavior must you consider at all times?

1. The person being interviewed will tell you what he or she views is in his or her best interests.
2. The person being interviewed can be subpoenaed to provide you any information you think he or she is hiding.
3. The person being interviewed will lie to you if you hit upon critical information.
4. The person being interviewed will only cooperate in the presence of an attorney or other representative.

Ans(1).Though the other items may be true from time to time, when conducting an incident investigation, remember that the person being interviewed will likely describe things that he or she perceives is in her best interest to discuss. This means that if the person being interviewed believes that the incident investigation is in his or her best interests, the interviewer will likely obtain the information sought. If the person being interviewed believes that job reprisals, legal action, or other adverse things will occur, the quality of the information obtained during an interview will likely be degraded.

Q271 Which of the following is the BEST method for conducting interviews associated with an incident investigation?

1. One interviewer and one witness in a room with audio recording
2. Three interviewers and one witness in a room
3. One interviewer and three witnesses in a room
4. One interviewer and one witness in a room

Ans(4). Audio and video recording should always be avoided as this will typically make the witness nervous and less likely to be fully open. Having three interviewers is intimidating to the witness. Interviewing multiple witnesses at the same time should be avoided since frequently, a dominant person will speak for the group. The other witnesses with information will reason with themselves that other witnesses in the room have better information - particularly if the information is contradictory.

Q272 A chemical process uses cryogenic liquid ethylene and other additives to produce the product. A process hazard analysis (PHA) of this process determined that a potentially dangerous situation could occur if the liquid ethylene were aligned directly with the flare (the flare is not designed for liquid ethylene). This situation would occur if Automatic Valve AV-1 and Manual Valve MV-C were opened while liquid ethylene was flowing. The PHA team determined four options to correct this issue. Which is the most desirable corrective action?

1. Create and test a software interlock in the programmable logic controller to prevent AV-1 from opening when MV-C is open and liquid ethylene is flowing (according to Flow Controller FC-1).
2. Install a flare that can handle liquid ethylene at the flow rate expected when this situation occurs.
3. Invoke a strictly enforced administrative procedure to physically verify that MV-C is closed before the control room operator is allowed to open AV-1 when flowing liquid ethylene.
4. Replace AV-1 with a manual valve and require operators to strictly enforce an administrative procedure to physically verify that MV-C is closed before the operator is allowed to open the new manual valve when flowing liquid ethylene.

Ans(1).Though all of these corrective actions could work, the most desirable is to install a software interlock. A new flare would likely be an expensive solution, and administrative controls are always less desirable than engineering controls.

Q273 A chemical process uses a flammable material under pressure. The flammable material is pumped into a reactor vessel (a pressure vessel). The pressure vessel is designed in accordance with ASME requirements. As a part of the process control, the safety relief valve is used to regulate pressure. When the vessel pressure is too high, the safety relief valve lifts, and a flammable mixture flows to the flare (the flare is designed to handle substantially more load than this). Which of the following statements best describes this process?

1. The process is designed adequately.
2. The process is not designed adequately because flammable materials should not be used under pressure.
3. The process is not designed adequately because safety relief valves should not be used to control the pressure in any process.
4. The process is not designed adequately because the products of the reactor should not be relieved to the flare through a safety relief valve.

Ans(3).It is considered poor engineering practice to control a process with its safety systems. Other process controls and design features should be put in place so that the safety system can be used to relieve the system when safety parameters have been exceeded.

Q274 A quantitative risk analysis (QRA) was performed on a single process. The results of the analysis indicate that the process will fail causing catastrophic consequences once every 10,000 years. Your plant has seven of these processes operating at all times, and your company has 12 other plants similar to this one around the country. Also, your company has three other competitors with similar numbers of facilities with very similar processes. Assuming that the results of the QRA can be extrapolated to the other similar processes throughout the industry, how often should you expect a catastrophic accident in your company's industry related to this process?

1. about once every 30 years
2. about once every 80 years
3. about once every 150 years
4. about once every 180 years

Ans(1). The results of the QRA are associated with a single process. Your company has 84 such processes (12 facilities * 7 processes/facility). Your company's competitors bring another 252 such processes (84 processes * 3). This is a total of 336 such processes in the industry. That means we can expect 336 catastrophic accidents in 10,000 years, or about one accident every 30 years.

Q275 Of the following, which is a reason for avoiding quantitative analyses?

1. Probabilities calculated are often so small that many believe corrective action need not be taken
2. Quantitative analyses are often expensive and frequently require a team of trained professionals to perform the analyses
3. Quantitative analysis techniques often require the use of personnel capable in mathematical techniques and/or use of a computer
4. All of the above

Ans(4). All these and more are reasons that quantitative analyses are not always accepted or done on product evaluations.

Q276 The deductive technique that selects the undesired outcome (top-level event) and all possible modes of happenings is the:

1. Failure modes effects analysis
2. Fault tree analysis
3. Fault hazard analysis
4. THERP

Ans(2). In the fault tree method, an undesired event is selected and all possible happenings that can contribute to the event are diagrammed in the form of a tree. The branches of the tree are continued until independent events are reached. Probabilities are determined for the independent events and after simplifying the tree, both the probability of the undesired event and the most likely chain of events leading up to it can be computed.

Q277 In developing a safe maintenance guide for robotic workstations, which method, useful for predicting potential injury situations, would apply to determining the combinations of human actions and equipment conditions that could lead to a robot related injury?

1. SADT
2. JSA
3. FTA
4. FMECA

Ans(3). Fault tree analysis (FTA) begins with defining the unwanted injury event and proceeds by graphically constructing the sequences of events and conditions that would lead to that event. When failure rates and human reliability values are available, the probabilities of the various sequences can be computed. NIOSH Publication No 88-108 Safe Maintenance Guide for Robotic Workstations SADT - Structural Analysis and Design Techniques JSA - Job Safety Analysis FMECA - Failure Modes, Effects and Critical Analysis

Q278 Given that incident investigations require resources (time and money), what would be an efficient method to maximize your company's ability to harvest as much information from each incident investigation that is conducted?

1. Perform a formal incident investigation on every incident regardless of complexity and consequence.
2. Perform a formal incident investigation on every complex incident and on every incident with a high consequence.
3. Trend all incidents, perform a formal incident investigation on every complex incident, on every incident with a high consequence, and on other incidents that show a trend.
4. Perform a formal investigation on a randomly selected set of incidents.

Ans(3).The key is trending. An incident can be as minor as a process upset leading to 5 seconds of downtime to a safety-related fatality due to a highly complex sequence of events. No company can afford to investigate every incident unless there is a method for choosing the incidents that will yield the most information and the most opportunity for improving. In some cases, regardless of the complexity, a formal investigation will be commissioned. More frequently, however, a company will want to trend incidents and only perform a formal incident investigation when the trend indicates that there is a higher-than-acceptable potential for a repeat. Trending applies to near misses, as well.

279 What is a chain of custody?

1. A documentation trail describing a detailed history of specific items particularily with respect to who has held custody of the item and the reason that person had custody of the item
2. A documentation trail describing the manufacturing history or personal history of an item or person
3. A documentation trail describing the quality review and checklists that were performed on an item
4. None of the above

Ans(1). A chain of custody is typically associated with incident investigations and describes in detail the identity of the person who has held custody of an item and why the person had custody of the item.

Q280 Change analysis is:

1. A method for understanding how an incident occurred
2. A method for understanding how a change in a system could cause problems later on
3. A method of understanding how a management system change will affect the system
4. All of the above

Ans(4).Change analysis is an excellent investigative tool for incident investigation as well as a proactive tool for performing risk assessments when a system change is anticipated.

Q281 The F-statistic is associated with problems involving:

1. ANOVA and inferences about standard deviations
2. Chi-Square distributions
3. Inferences about means
4. None of the above

Ans(1). When performing an analysis of variance test (ANOVA) the F-statistic is computed. Chi-Square tests involve the Chi-Square distribution. Inferences about means usually involve the assumption of normal distributions for large samples and the "t" distribution for small samples.

Q282 Which of the following is NOT generally considered part of a behavioral observation process?

1. Identification of behaviors to observe
2. Engineering controls
3. Conduct of observations
4. Provide feedback on results

Ans(2). A behavioral observation process generally consists of identifying the behaviors to be observed, conducting the observations, and providing feedback on the results.

E. Scott Geller, The Psychology of Safety

Q283 Extinction can be accomplished by:

1. Punishment
2. Withholding positive reinforcement
3. Both 1 and 2
4. None of the above

Ans(3). Extinction can be accomplished by punishment and withholding positive reinforcement.

Dan Petersen, Safety Management: A Human Approach

Q284 A Theory Y leader would exhibit which type of management style?

1. Autocratic
2. Democratic
3. Supportive
4. Consultative

Ans(2).Theory Y leaders assume that their employees are intelligent, hard working, and willing to learn. They would encourage subordinates to self-manage, take responsibility, and share in decision making. This is democratic leadership.

Q285 To assess your company's loss control performance, several statistics are captured and analyzed. One of these is "Percent of operational personnel that have completed all phases of chemical safety training." This statistic is an example of a(n):

1. Leading indicator
2. Lagging indicator
3. Optional indicator
4. Occupational indicator

Ans(1).Leading indicators (such as risk assessments) provide data on how well the loss control system is expected to perform in the future. In this particular case, there is evidence that when a lower fraction of operational personnel have been trained, the company's loss control performance decreases.

Q286 Calculate the standard deviation of the following sample set of data: 52, 20, 24, 31, 35, 42.

1. 10.8
2. 11.8
3. 12.8
4. 13.8

Ans(2).Most scientific calculators will calculate the population standard deviation (it may be labeled as σn). The formula used to calculate the sample standard deviation is

$$s = \sqrt{\frac{\Sigma (x - \overline{x})^2}{n - 1}}$$

where x is the difference between the sample data and the sample mean and n is the number of sample data points. The sample standard deviation is 11.8

Q287 Estimate the variance of the population from which the following sample data came: 22, 18, 17, 20, 21.

1. 4.3
2. 2.1
3. 1.9
4. 5.0

Ans(1).Use the following formula to calculate the sample standard deviation:

$$s = \sqrt{\frac{\Sigma (x - \overline{x})^2}{n - 1}}$$

The variance is then calculated by squaring the sample standard deviation. As we have no other information to draw on, we use the sample variance as a point estimate of the population variance.
The sample standard deviation is 2.074
$2.074^2 = 4.3$

Q288 What is the best estimate of the variance of the population from which the following samples came? 17 ppm, 20 ppm, 18 ppm, 22 ppm, 21 ppm, 22 ppm, 19 ppm, 21 ppm, 22 ppm

1. 3.44
2. 4.30
3. 5.01
4. 5.13

Ans(1).The best estimate of the population variance is the sample variance (the sample standard deviation squared). To calculate the sample standard deviation, use the formula:

$$s = \sqrt{\frac{\Sigma (x - \overline{x})^2}{n - 1}}$$

The sample standard deviation = 1.856
$1.856^2 = 3.44$

Q289 A normally distributed population has total of 150 members and a standard deviation of 24. What is the standard error of the mean for a sample from this population?

1. 0.16
2. 1.60
3. 4.90
4. Cannot be determined

Ans(4). A sample size is not given and hence a question about the standard error of the mean is not relevant. The standard error of the mean is determined by dividing the population standard deviation by the square root of the sample size.

Q290 The hidden costs of accidents are:

1. 1 to 4 times the direct costs
2. 4 to 10 times the direct costs
3. 10 to 20 times the direct costs
4. 50% of the direct costs

Ans(2). There are direct and indirect costs associated with an accident. These costs can be thought of as an iceberg, with the direct costs the visible portion of the iceberg and the indirect costs the large ice mass below the surface of the water. The indirect costs have been estimated to be 4 to 10 times the direct costs.

Q291 Cost factor equals the total compensation and medical costs incurred per exposure, and is calculated by the following:

1. CF = Exposure/cost incurred
2. CF = Cost * exposure
3. CF = Cost incurred * 1,000/total work-hours
4. CF = Total work-hours/cost incurred * 100,000

Ans(3). Cost incurred * 1,000 / total work-hours is the formula for cost factor.

Q292 The manual premium for a company under workers' compensation law is:

1. A premium from a manual for that type of company
2. Premium modified for the company's experience
3. The premium before experience modification is applied
4. The premium before automatic debits and credits are applied for safety program actions the company has taken

Ans(3). The manual premium is the premium before experience modification is applied. Modified premium is manual premium modified by a company's specific experience modifier It is also known as the standard premium. If any credits or debits are applied for specific program actions, they are usually applied after experience modification is applied.

Q293 If a company is an assigned risk under a workers' compensation program, it means that the company:

1. Has been assigned a classification for the degree of risk it presents to its employees
2. Is unable to secure coverage and has been assigned to an insurance company for workers' compensation coverage
3. Has been assigned to a group that represents the type of products or services it produces
4. Is a good risk, has low losses, and therefore is assigned a preferred rate

Ans(2). Assigned risks are also know as pool risks, or residual market risks. Most assigned risks either have higher than anticipated losses, are new companies without any history of losses to predict their actual losses, or are in financial difficulty, making them less desirable to insure.

Q294 In workers' compensation systems, the premium a company pays for coverage is a function of:

1. Payroll
2. Manual class rates
3. Experience modification
4. All of the above

Ans(4). Premium is determined by Payroll/$100 * Class Rate * Experience Modification.

Q295 Which of the following traditional insurance coverages for business is a "no-fault" coverage?

1. Automobile insurance
2. Product liability insurance
3. Premises liability insurance
4. Workers' compensation insurance

Ans(4). Workers' compensation insurance is one of the first types of "no-fault" coverages for US businesses. No-fault coverage means that the injured party is compensated no matter who is at fault. While some states have sought to bring the concept of no-fault to other coverages such as automobile coverage, workers' compensation remains one of the only types of insurance that began as a no-fault coverage.

Q296 In workers' compensation, most insurance is provided by private sector insurance companies. When insurance can only be obtained from the state government, that state is known as:

1. A residual market state
2. A state plan state
3. A state fund state
4. A pool state

Ans(3). States that operate their own insurance mechanism for workers' compensation are known as state fund states.

Q297 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, what did this company pay for worker's compensation insurance in 1998?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. $181,440
2. $208,656
3. $242,987
4. $332,435

Ans(2). Premium = (Payroll/$100) * Rate * Experience Modification = ($1,680,000/$100) * $10.80 * 1.15 = $208,656

Q298 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, if this company had losses of $245,000 in 1997, what was its loss ratio?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. 78%
2. 85%
3. 117%
4. 129%

Ans(4). Loss Ratio = Losses / (payroll/$100) * Rate * Experience Modification

LR = $245,000 / [($1,550,000 / $100) * $11.15 * 1.10]
= $245,000 / $190,107.50
= 1.29 or 129%

Q299 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, in which year or years of experience would this company's experience modifier for the year 2000 be based?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. 1998
2. 1997, 1998, 1999
3. 1996, 1997, 1998
4. 1999

Ans(3). The experience modifier for a company is based on the last three years of actual experience, not counting the most recent year.

Q300 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, what does the experience modifier of 1.00 in 1996 indicate?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. The company had average losses in that year
2. The company was too new and did not have enough past actual experience to have a derived experience
3. The company had average losses in 1995
4. The company had changed their worker's compensation Class in that year because they changed the product they

Ans(2). The experience modifier of 1.00 most likely indicated that the company was too new to have its own derived experience modifier. It could also indicate an ownership change or that the firm had average losses (experience modifier = 1.00) for several years in a row.

Q301 A company manufactures machined parts and is in a worker's compensation Class for machine shops. For the year 2000, the test modifier is 1.16. Based on this information and the

data provided in the Table below, if this company changes the product it makes to furniture in the year 2000, with a worker's compensation Class rate of $6.00, what will its premium be per $100 of payroll?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. $6.00
2. $6.96
3. $11.00
4. $12.76

Ans(2). The rate per $100 of payroll is the test experience modifier times the new worker's compensation class rate. The calculation retains the company's experience modifier.

Rate = Test Modifier * New worker's compensation Class Rate
= 1.16 * $6.00
= $6.96

Q302 A company manufactures machined parts and is in a worker's compensation Class for machine shops. For the year 2000, the test modifier is 1.16. Based on this information and the data provided in the Table below, if this company is sold to another group of owners who change the product they make in 1997 to furniture, with a worker's compensation Class rate of $6.00, what will its premium be per $100 of payroll?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. $6.00
2. $6.96
3. $11.00
4. $12.76

Ans(1). When ownership changes the experience modification reverts back to 1.00. Therefore, the class rate per $100 of payroll would be the manual rate for that class of business. In this case the class has changed because the product made has changed.

Q303 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, which of the following statements is TRUE?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. This company's losses are trending down
2. We do not have sufficient information to determine what is happening with loss trends
3. This company has losses better than other companies in this industry
4. This company's losses are trending up

Ans(4). Experience modifiers, when compared from year to year is an indicator of the trend of losses. It is a lagging indicator since it is an average of three years of losses and expected losses.

Q304 A company manufactures machined parts and is in a worker's compensation Class for machine shops. For the year 2000, the test modifier is 1.16. Based on this information and the data provided in the Table below, when this company's management decided to build a new plant in another state in 1997 to manufacture furniture, what experience modifier was used for the new plant?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000

1. Not enough information is provided
2. 1.00
3. 1.10
4. 1.15

Ans(3). The experience modifier is an expression of the experience of the firm's managers/owners. Even though a new plant is built in another state, the new plant takes on the experience modifier of the management/owners of the existing company. The class rate/manual rate changes in this case, but the experience modifier remains the same.

Q305 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on data provided in the Table below, and assuming that the ratable losses and expected losses in 1999 did not change significantly from the previous years, what is the firm's estimated modifier for the year 2000?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Total Ratable Losses	$375,580	$375,580	$375,580	
Total Expected Losses	$284,530	$284,530	$284,530	

1. Not enough information is provided
2. 0.76
3. 1.16
4. 1.32

Ans(4). Each state sets its own formula to calculate the experience modifier based on the ratio of ratable (actual) losses with expected losses for the workers compensation Class. Some states average only the last year's ratio while other states permit the averaging of the last several years ratios.

Experience modifiers = ratable losses / expected losses
EM = $375,580 / $284,430
= 1.32

Q306 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on data provided in the Table below, what is the OSHA incident rate for cases in 1996?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Work Hours	275,000	286,000	290,000	310,000
Number of Accidents	90	88	96	101
Lost Work Day Cases	12	12	9	7

1. 0.087 cases/year
2. 0.65 cases/year
3. 65 cases/year
4. 87 cases/year

Ans(3).Rate = No. of Cases (200,000 hr/yr) / (Total Work Hours)
= 90 cases(200,000 hr/yr) / 275,000 hr
= 65.45

Remember that the OSHA incidence rate requires basing the calculation on 100 employees each working 2,000 hours per year (that is the purpose of the 200,000 hr/yr in the above equation).

Q307 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on data provided in the Table below, what is the OSHA lost work day incidence rate for cases in 1999?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Work Hours	275,000	286,000	290,000	310,000
Number of Accidents	90	88	96	101
Lost Work Day Cases	12	12	9	7

1. 1.5
2. 4.5
3. 15
4. 45

Ans(2). Rate = No. of Cases (200,000 hr/yr) / (Total Work Hours)
= 7 cases (200,000 hr/yr) / 310,000 hr
= 4.5

Remember that the OSHA incidence rate requires basing the calculation on 100 employees each working 2,000 hours per year (that is the purpose of the 200,000 hr/yr in the above equation).

Q308 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on incidence rates (see Table below), what is the trend of total accidents for this company?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Work Hours	275,000	286,000	290,000	310,000
Number of Accidents	90	88	96	101
Lost Work Day Cases	12	12	9	7

1. Accident rates are increasing
2. Accident rates are decreasing
3. Not enough information is provided
4. The trend is not apparent since the change from year to year is small

Ans(4). The incidence rates for 1996 through 1999 were 65, 61.5, 66.2, and 65, respectively. There is not a great change from year to year to indicate any definite trend.

NOTE: Incidence rate is calculated by:
Rate = No. of Cases(200,000 hr/yr) / (Total Work Hours)

Q309 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on incidence rates (see Table below), what is the trend of lost work day accidents for this company?

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Work Hours	275,000	286,000	290,000	310,000
Number of Accidents	90	88	96	101
Lost Work Day Cases	12	12	9	7

1. Accident rates are increasing

2. Accident rates are decreasing
3. Not enough information is provided
4. The trend is not apparent since the change from year to year is too small

Ans(2). The incidence rates for 1996 through 1999 were 8.7, 8.4, 6.2, and 4.5. This is clearly a downward trend.

NOTE: Incidence rate is calculated by:
Rate = No. of Cases (200,000 hr/year) / (Total Work Hours)

Q310 A company manufactures machined parts and is in a worker's compensation Class for machine shops. Based on the data provided in the Table below, which of the following is the best indicator of how this company's safety program is working.

	1996	1997	1998	1999
Class Rates	$10.90	$11.15	$10.80	$11.00
Experience Modifiers	1.00	1.10	1.15	1.22
Payroll $	$1,330,000	$1,550,000	$1,680,000	$2,225,000
Work Hours	275,000	286,000	290,000	310,000
Accidents	90	88	96	101
Lost Work Day Cases	12	12	9	7

1. Experience modifier
2. Test modifier
3. OSHA incidence rate for all fatality cases
4. OSHA incidence rate for all lost work day cases

Ans(4). The OSHA incidence rate for lost work day cases is the best indicator. Experience modifiers and test modifiers are also indicators but they are based on past losses over three years; the OSHA incidence rate is more current.

The fatality incidence rate will rarely have sufficient data from which trends can be drawn.

Q311 The preferred course for a company's management to meet the consequent risks of product liability is:

1. Regulation of risks through careful statement of conditions of sale and delivery
2. Transfer of risks through the underwriting of product liability insurance
3. Reduction of risks through the introduction of effective quality assurance
4. Acceptance of risks through passivity

Ans(3). The management of a company must consider what ways or combinations of ways are available to meet the consequent risks of product liability. Of the four fundamentally

different risk alternatives that the company's management may take, the reduction of the existing risk through introduction or improvement of quality assurance will give better conditions for regulation and transfer of risk.

Q312 The manufacturer of any product faces an environment in which:

1. Fault without negligence exists
2. Present court rulings favor the company
3. Product liability is a not a significant financial burden
4. Product liability claims are generally decreasing

Ans(1). The manufacturer of any product faces an environment in which fault without negligence exists. Legal history shows a movement from buyer beware to seller beware.

Q313 Alpha Company sold a product in 1990 that caused an accident in 1999. The company was not insured in 1990, but did acquire Mutual B product liability insurance for the same product in 1992. Which firm is responsible for losses?

1. Alpha Company since it was not insured in 1990
2. Mutual B
3. Both Alpha and Mutual B share responsibility
4. Neither is responsible

Ans(2). Mutual B is likely still responsible for the losses unless specific disclaimers in the policy allow Mutual B to remain clear of product liability for products sold prior to the underwriting.

Q314 A tool made by Echo Corporation is sold to Tango Corporation by Wholesale Company. A Tango employee modifies the Echo tool and is subsequently injured using the employee-modified tool. Which firm is potentially liable for damages?

1. Echo
2. Tango
3. Wholesale
4. All parties are potentially liable

Ans(4). All parties involved are potentially liable in this situation. Tango's liability will likely not be associated with product liability, but biased toward worker-related injuries (if the employee elects to reject a workers' compensation settlement).

Q315 For potential legal protection, how long should a manufacturer keep product records?

1. Life of the product
2. Life of the product guarantee
3. Well beyond life of the product
4. Should not keep any records

Ans(3). Due to product liability claims, a manufacturer should keep product records well beyond the intended life of the product.

Q316 Implied warranty involves:

1. A written statement
2. Expectation of suitability
3. Negligence
4. Tort

Ans(2). An implied warranty revolves around the implications, made by a manufacturer or dealer, that a product is suitable for a particular use.

Q317 Which of the following is least likely to support a product liability claim?

1. Defective design
2. Defective manufacturing processes
3. Poor product instructions
4. Misuse by the consumer

Ans(4). Even customer misuse may be used as valid support for a product liability claim if the misuse could have been reasonably anticipated by the manufacturer; however, the other three answers are much easier foundations on which to justify a product liability claim.

Q318 The risk value for a given hazardous event is a measurement that combines all of the following variables EXCEPT:

1. Frequency of process
2. Unanticipated loss
3. Exposed systems
4. Likelihood

Ans(2). Unanticipated loss is not a variable for the risk value of a given hazardous event.

Q319 An insurable pure risk is NOT which of the following?

1. A car accident
2. A burglary
3. A business failure
4. A death

Ans(3). A business failure does not meet the definition of a pure risk since it is not accidental. It is also a single exposure and an event unique to that business.

Q320 Dealing with the uncertainty and unpredictability of personal injury, property loss, business interruption, and liability is identified with:

1. Casualty insurance
2. Loss prevention
3. Risk management
4. Loss control management

Ans(3). Risk management is a decision-making process whose objective is to identify the proper mix of loss control, risk retention, and risk transfer that the organization should use in dealing with its risks of personal injury, property loss, and liability, including the expense and revenue losses associated with these exposures.

Q321 Under workers' compensation laws:

1. The employee has to prove negligence by the employer
2. The amount of compensation is decided by the employee and employer under arbitration
3. The employer assumes responsibility and compensation for accidents, regardless of fault
4. The employee may sue the employer if the compensation provided is not adequate

Ans(3). Under workers' compensation laws, the employer assumes responsibility for all accidents and compensation to the employee, regardless of fault. The employee does not have to prove negligence and is not allowed to sue for damages beyond what is set by the state's workers' compensation law.

Q322 Two valves are in series. Valve "A" has failure rate of 1×10^{-3}. Valve "B" has a failure rate of 1×10^{-2}. What is failure rate of the system?

1. 1×10^{-2}
2. 1×10^{-3}
3. 1×10^{-5}
4. 1×10^{-10}

Ans(1). Since the valves are in series, you must add the failure rates.
Failure Rate = $1 \times 10^{-2} + 1 \times 10^{-3}$
= 1.1×10^{-2}

Q323 A boiler has three valves in series that have failure rates of:
3×10^{-3}
8×10^{-4}
2×10^{-4}

What is the overall failure rate of the system?

1. 48×10^{-11}
2. 13×10^{-11}
3. 48×10^{-3}
4. 4×10^{-3}

Ans(4).Add the failure rates of components in series.
Failure Rate = $3 \times 10^{-3} + 8 \times 10^{-4} + 2 \times 10^{-4}$
= 4×10^{-3}

Q324 An airplane landing gear has 4 components in parallel with failure rates of 1×10^{-5}, 1×10^{-8}, 1×10^{-8}, and 1×10^{-9}, respectively. What is the approximate failure rate?

1. 1×10^{-37}
2. 1×10^{-30}
3. 4×10^{-8}
4. 3×10^{-8}

Ans(2). Multiply the failure rates for parallel construction.
Failure Rate = $(1 \times 10^{-5})\ (1 \times 10^{-8})(1 \times 10^{-8})(1 \times 10^{-9}) = 1 \times 10^{-30}$

Q325 The heart of the systems safety approach is the:

1. Failure analysis
2. Hazard analysis
3. Charting and modeling
4. Structural analysis

Ans(2). Hazard analysis is the heart of the system safety approach. An effective hazard analysis effort over the life cycle of a system is the spine on which all body components of a safety program are attached. Anticipating and controlling hazards at the design stages of an activity is the cornerstone of a system safety effort.
"System Safety Engineering and Management" by Harold E. Roland and Brian Moriarity

Q326 A product safety factor is defined as:

1. The extra margin of safety over intended use
2. The product "Safeness"
3. The product's cost factor override
4. Amount of product liability insurance needed

Ans(1). The product safety factor is that extra margin of safety designed into the product over and above its intended use.

Q327 All of the following are examples of redundant design philosophy EXCEPT:

1. Series
2. Single parallel
3. Double parallel
4. Boolean design

Ans(4). Series, single parallel, and double parallel are all examples of redundant design philosophy.

Q328 What is critically needed prior to conducting a formal system safety analysis?

1. System models
2. Management cooperation
3. Detailed engineering drawings
4. Liability transfer

Ans(1). A system safety analysis must have models of the involved system. Before analysis of safety can be made, a model of the system must be constructed.

Q329 To increase the reliability of a process, which of the following methods would LEAST likely be used?

1. Design parallel components
2. Design complementary components
3. Design redundancy into the system
4. Design series components

Ans(2). Parallel and series design, as well as redundancy, are all appropriate methods for increasing the reliability of a system.

Q330 What is a bathtub curve?

1. New product innovation cycle
2. Failure rate analysis showing a spiraling rate of failure
3. Typical failure rate over the life of a product
4. Atypical failure rate over the life of a product

Ans(3). The bathtub curve traces the typical failure rate of a product over time.

Q331 A procedure analysis is a product evaluation technique that is similar to:

1. Accident investigation.
2. Job safety analysis
3. Failure modes and effects analysis
4. Preliminary hazard analysis

Ans(2). The techniques of a procedure analysis are similar to those of a job safety analysis except that in a JSA one identifies the actions the employee might take that could injure himself/herself, injure a fellow employee, or do damage to the operations. In a procedure analysis, the evaluator looks for procedures that the employee might use in the manufacturing of a product that could induce a hazard in the product.

Q332 What is a CMMS?

1. A strategy for managing occupational safety in a construction environment
2. A computerized data management system for tracking equipment maintenance tasks
3. A computerized chemical safety management system
4. A method for managing compliance with Minerals Management Service regulations

Ans(2).CMMS stands for computerized maintenance management system. It is designed to assist maintenance and operations professionals with tracking equipment maintenance tasks and equipment history.

Q333 What is the fundamental objective that a developer of a form, report, or other written instrument must remember?

1. The needs of the end user of the written instrument
2. The needs of the one who will read the written instrument
3. The needs of the ones described by the written instrument
4. All of the above

Ans(4). A developer must balance these three needs when creating forms, reports, and other written instrument.

Q334 There are several safety-related uses for personal computers. Considering the vast amount of information relating to safety, computers should allow safety practitioners, employees, and supervisors access to anything to mitigate risks. Of the following, what is one of the most critical pieces of safety-related information that company employees should have computer access to?

1. OSHA-reportable/recordable injury and illness statistics
2. Near miss/lessons learned data
3. Company occupational accident rates, by plant
4. Company occupational illness rates, by department

Ans(2). Of these, the information that is extremely helpful is the kind which helps in PREVENTING future accidents - near miss and lessons learned data. Unfortunately, most companies do a very poor job of tracking this type of information, let alone granting employees computer access to it.

The other information is somewhat helpful for employees to have on a computer; however, real-time access to historical data is only marginally useful in preventing future accidents.

Q335 Heinrich's ratio of no-injury accidents to minor injuries to major injuries is:

1. 100-20-1
2. 200-30-10
3. 300-29-1
4. 600-58-1

Ans(3). Heinrich wrote that for every 300 no-injury accidents, there are 29 minor injuries accidents and 1 major injury accident.

Q336 In 1931, Heinrich significantly changed the thinking of safety professionals. He:

1. Shifted emphasis from conditions to actions
2. Suggested the use of JSA to analyze accidents
3. Applied forensic science to accident analysis
4. Proposed monetary penalties for accidents

Ans(1). In the early 1900's, employers concentrated more on the inspection of the workplace. Heinrich suggested that more emphasis be placed on the actions of employees.

Q337 It is generally accepted that accidents:

1. Are all preventable
2. Are preventable about 50% of the time
3. Are unexpected events that cannot be controlled
4. Can generally be prevented, but where absolute prevention is not possible, steps can be taken to reduce accident severity

Ans(4). Each of the answers is a history of accident prevention. The prevalent concept in the early 1900's was that no more than 50% of all accidents were preventable. In the 1940's, it changed to a concept that all were preventable, and finally to today's more realistic theory that not all accidents can be prevented, but in those cases where absolute prevention is not possible, steps can be taken to reduce the severity of the damage or injury.

Q338 Your company's chief executive officer feels that she has too many employees reporting directly to her (among whom is you, the safety director). She has asked your opinion about where in the organizational structure is the best place from which the safety function should report? Your opinion is:

1. Safety should report through Human Resources
2. Safety should report through Procurement Support
3. Safety should report Production Operations
4. Safety should remain on the chief executive officer's staff

Ans(4). Safety should report to a function with the authority and the control to correct company-wide safety problems. Therefore, the safety function should continue reporting to the chief executive officer.

Q339 Until the widespread availability of inexpensive personal computers, the information associated with safety-related facility and site inspections and surveys was difficult to maintain. Of the following, what is the best use of the personal computer to manage information from these endeavors?

1. Organizing and categorizing findings
2. Maintaining an up-to-date list of responsible supervisors
3. Tracking and following up on the results of recommendations and corrective actions
4. Allowing for consistent documentation and reports

Ans(3). Though all of these have benefits, the key advantage to using personal computers is the ability to track and verify that recommendations and corrective actions are being addressed in a timely fashion. The key purpose of inspections and surveys is to prevent accidents, and the implementation of these corrective actions and recommendations is management's window to verifying that these identified risks have been addressed.

Q340 The following clause from the OSH Act of 1970 is known as: "...of employment which are free from recognized hazards that are causing or likely to cause death or serious physical harm to his employees"

1. "Recognized hazard clause"
2. "General duty clause"
3. "General hazard clause"
4. "Death or injury clause"

Ans(2).This clause gives OSHA broad regulatory license. It is known as the "general duty clause". If there is a serious hazard for which OSHA does not have a specific standard, and OSHA compliance officer can cite the employer using the General Duty Clause, thus requiring the hazard to be abated.

Q341 At one time, a plaintiff could not recover damages unless the injury was due to physical contact. This principle is called:

1. Foreseeability
2. Negligence per se
3. Fright without physical contact
4. Proximate cause

Ans(3). Fright without physical contact is the principle that a plaintiff could not recover damages unless the injury was due to physical contact.

Q342 When an injured person's care for his own safety was less than reasonable for a prudent man under existing conditions, he is considered negligent and the defendant will not be held liable. This principle is called:

1. Gross negligence
2. Contributory negligence
3. Comparative negligence
4. Strict liability

Ans(2). Contributory negligence is the principle that an injured person's care for his own safety was less than reasonable for a prudent man under existing conditions. He is considered negligent and the defendant will not be held liable.

Q343 Every person has a legal duty to exercise due care for the safety of others and to avoid injury to others if possible. This principle is called:

1. Exercise of due care
2. Slight care
3. Non gross negligence
4. Great care

Ans(1). Exercise of due care is the principle that every person has a legal duty to exercise due care for the safety of others and to avoid injury to others if possible.

Q344 A standard applying to a particular industry with specifications that relate to individual operations is termed a:

1. Design standard
2. Performance standard
3. Vertical standard
4. Horizontal standard

Ans(3). A vertical standard pertains only to a particular industry (e.g., OSHA requirements for pulp, paper, and paperboard mills).

Q345 Which of the following statements about regulatory standards is INCORRECT?

1. Regulatory standard are developed by a federal agency such as OSHA
2. Regulatory standards can be developed by state level agencies
3. Regulatory standards are enforceable
4. Regulatory standards are voluntary

Ans(4). Regulatory standards are mandatory. If voluntary, they are not standards.

Q346 The "Whirlpool" decision of the U.S. Supreme Court dealt with:

1. Employee right to refuse life-threatening work
2. Employee right to refuse a respirator
3. Employee right to refuse medical exam
4. Employee right to request an OSHA inspection

Ans(1). The "Whirlpool" decision established an employee's right to refuse life- threatening work.

Q347 The first legally upheld workers' compensation law was in which state:

1. Minnesota
2. Wisconsin
3. Massachusetts
4. Alabama

Ans(2).Wisconsin had the first workers' compensation law.

Q348 The first state to utilize factory inspectors was:

1. Massachusetts
2. Illinois
3. Minnesota
4. Wisconsin

Ans(1). Massachusetts was the first state to have factory inspectors in 1867.

Q349 The most dangerous industry from the 19th to the 20th century was:

1. Automobile manufacturing
2. Mining
3. Railroad
4. Construction

Ans(2). Mining has been (and still is) one of the most dangerous industries in the United States. Fire fighting, construction, and logging also remain high on the list.

Q350 What was one of the first incentives for management to ensure safe working conditions for employees?

1. The Occupational Safety and Health Act
2. Workers' compensation legislation
3. Reduction of insurance premiums
4. Increased productivity

Ans(2).In 1911, the Wisconsin workers' compensation law was the first law of its kind that required compensation for injuries which occurred on the job, without regard to fault.

Q351 Common law requires employers to:

1. Provide a workplace free of hazards
2. Compensate employees for workplace injuries
3. Provide a workplace that is reasonably safe
4. Applies to employees NOT employers

Ans(3). Common law only requires the employer to provide a workplace that is reasonably safe.

Q352 The standard error of the mean:

1. Is always less than the population standard deviation
2. Cannot be determined if the population standard deviation is not known
3. Increases as the sample size increases
4. None of the above

Ans(4). In general, the standard error of the mean is always less than the population standard deviation. However, if the sample size is 1, the standard error of the mean is the standard deviation.

The standard error of the mean cannot be determined if the population standard deviation is not known is not true because the standard error of the mean can be determined by using a point estimate derived from a sample standard deviation. The standard error of the mean decreases as the sample size increases.

Q353 When dealing with outliers in statistical data sets one should:

1. Always eliminate them from consideration since it is obvious that a mistake has been made
2. Be suspect but always give careful consideration before discarding them
3. Rerun the experiment or take a new sample
4. Include them in the analysis since there is no way to determine how they appear

Ans(2). There are many sources of outliers including instrument failure, errors in reading instruments, poor experimental design, etc. But it is always possible there is something going on that would legitimately produce that result. Since they can have a considerable influence on the data set they should always be carefully considered before being accepted or summarily discarded.

Q354 Inferential statistics can best be described as:

1. Setting up an experimental design for collecting data to be analyzed
2. Organizing, presenting, and analyzing data
3. Making predictions from data
4. Predicting the probabilities of certain events occurring in a population

Ans(4). Predicting the probabilities of certain events occurring in a population is a basic definition of inferential statistics. We are concerned with predicting or inferring something about a population usually based on a sample.

Q355 Which of the following measures is likely to be the best measure of central tendency in a skewed (or non-symmetric) distribution?

1. Mean
2. Median
3. Mode
4. They are all affected equally

Ans(2).A measure of central tendency is a number which represents the "average" value for a set of scores. The MEAN is computed by summing all the scores and dividing by the number of scores. The MODE is the most frequently-occurring score in a distribution. The MEDIAN is computed by rank-ordering all the scores and taking the one which is sequentially in middle. When a distribution of scores is "normal" and not skewed, the mean, median and mode are all the same score. When the distribution is skewed, they all differ. Under these conditions, it is generally accepted that the median is less affected than the other measures of central tendency.

Q356 What is the best definition regarding percentiles that defines the interquartile range?

1. The 95th percentile
2. The range from the 25th to the 75th percentiles
3. The 5th percentile
4. The "average range"

Ans(2). The 95th percentile is incorrect. This means that 95% of the subjects fall within this percentile.

The 5th percentile is incorrect. This means that 5% of the group falls within this number.

Q357 If variable 1 and variable 2 are highly correlated with one another (r = 1.0), this means:

1. Variable 1 causes variable 2
2. Variable 2 causes variable 1
3. Both 1 and 2
4. None of the above

Ans(4). Even if two variables are perfectly correlated with one another (r = 1.0), we cannot say that one CAUSES the other. For all we know, a third variable may be responsible for the observed correlation. A correlation coefficient simply indicates the degree to which two variables covary. As such, you can generate correlations between any two measurable things (e.g., the amount of rainfall in Burma with the number of baseball tickets sold in New York). Such correlations might not be interesting, but it is still possible to compute them even though they have no conceivable causal relationship to one another.

Q358 If two variables (1 and 2) have a correlation of r = -0.09, this would be considered a:

1. Strong, direct correlation
2. Weak, direct correlation
3. Strong, inverted correlation
4. Weak, inverted correlation

Ans(4).Correlation coefficients can range from -1.0 to 1.0. The negative value indicates that the correlation is inverted. The correlation of -0.09 indicates that the relationship between the two variables is very weak.

Q359 The mode is defined as:

1. The average of all the numbers in a sequence
2. The value which appears most frequently in a sequence
3. The difference between the highest and lowest values in a sequence
4. The value in a sequence for which there are an equal number of values greater than the value and less than the value

Ans(2).The mode is defined as the value in a sequence which appears most frequently. The median is the number in the middle of a sequence and average or mean is an arithmetic average of the sum of the numbers divided by the number (n) in the set.

Q360 The median is defined as:

1. The average of all the numbers in a sequence
2. The number which appears most frequently in a sequence of numbers
3. The difference between the highest and lowest values in a sequence
4. The value in a sequence for which there are an equal number of data points greater than the value and less than the value

Ans(4).The median is defined as the value in a sequence for which there are an equal number of data points greater than the value and less than the value. The mean is the arithmetic average of the set and the mode is the number in the set which appears most frequently.

Q361 The standard deviation is:

1. A measure of the average distance of all observations from the mean
2. A measure of the average distance of all observations from the median
3. A constant which accounts for randomness
4. Not useful when working with normal distributions
5.

Ans(1). The standard deviation is a measure of the average distance of all observations from the mean.

Q362 In a normal distribution, approximately 95.4% of the observations will fall between:

1. The median plus or minus one standard deviation
2. The median plus or minus two standard deviations
3. The mean plus or minus one standard deviation
4. The mean plus or minus two standard deviations

Ans(4). In a normal distribution, 95.4% of the observations will fall between the mean plus or minus two standard deviations.

Q363 In a normal distribution, about 68.3% of the observations will fall between:

1. The median plus or minus one standard deviation
2. The median plus or minus two standard deviations
3. The mean plus or minus one standard deviation
4. The mean plus or minus two standard deviations

Ans(3). In a normal distribution, about 68.3% of observations will fall between the mean plus or minus one standard deviation.

Q364 The geometric mean:

1. Is synonymous with the arithmetic mean
2. Is used when evaluating log normally distributed data
3. Is the median of log normally distributed data
4. Both 2 and 3

Ans(4). The geometric mean is used when evaluating log normally distributed data and is equal to the median of the log normally distributed data. The geometric mean is the square root of the product of all of the data points.

Q365 Which of the following is TRUE about a normal distribution?

1. It is symmetric around the mean
2. It can be characterized by the mean and the standard deviation
3. It is also known as a Gaussian distribution
4. All of the above

Ans(4). A normal, or Gaussian, distribution is symmetrical around the mean and is completely characterized by the mean and standard deviation.

Q366 What is the advantage of the use of historical control groups in toxicology studies?

1. No concurrent controls needed
2. Statistical value
3. Cost savings
4. There is no advantage

Ans(2). In the event of a non-statistically significant increase in a rare tumor type in treated animals, a low level or lack of occurrence in historical controls gives greater biological validity to the finding. Use of only a concurrent control group which is usually smaller might have resulted in dismissal of the effect as due to chance alone.

Q367 Of the following, the LEAST likely principle value of accident statistics is:

1. Measurement of performance
2. Determining trouble spots
3. OSHA reporting requirements
4. Lower worker's compensation costs

Ans(4). Accident statistics are used to measure performance, detect trouble spots, and fulfill OSHA record-keeping requirements.

Q368 Descriptive statistics can best be defined as statistics that:

1. Deal with methods of planning the observation
2. Summarize population data
3. Make generalizations based on sample data
4. Provide information about individuals rather than groups

Ans(2). Descriptive statistics is the branch of statistics that deals with methods that help to classify, summarize, and present the results of observations in forms that are comprehensible and likely to be suggestive of fruitful hypotheses.

Q369 What is the preferred method for measuring variability in statistical analysis?

1. Arithmetic mean
2. Standard deviation
3. Absolute dispersion
4. Mean square

Ans(2). The standard deviation is the most widely used measure of dispersion (or variability, or spread). The computation of the standard deviation is based on, and is representative of, the deviations of the individual observations about the mean of those values.

Q370 In Boolean logic and its application, the LEAST likely definition for a set is:

1. A group of objects having at least one characteristic in common
2. A collection of ideas having at least one characteristic in common
3. A collection of mathematical relationships having at least one characteristic in common
4. A group of objects having no characteristics in common

Ans(4). Boolean algebra was developed originally for the study of symbolic logic. Its rules and expressions in mathematical symbols permit complicated propositions to be clarified and simplified. In Boolean algebra, a set is a group of objects having at least one characteristic in common. The set may be a collection of objects, conditions, events, symbols, ideas, or mathematical relationships.

Q371 A process of examining any system, design, or operation to discover inherent hazards, characterizing them as to the level of risk, and identifying risk-reduction alternatives is called a:

1. Failure modes and effects analysis
2. Job safety analysis
3. Fault tree analysis
4. Hazard analysis

Ans(4). A hazard analysis is the process of examining a system, design, or operation to discover inherent hazards, characterizing them as to the level of risk, and identifying risk-reduction alternatives. Of course, there are several TYPES of hazard analysis techniques.

Q372 A successful research study is characterized by which of the following?

1. The study must include random samples
2. The study and control groups must show the same results
3. Variables for the study and control group must be controlled by one person
4. Evidence of the study must be reproducible by others

Ans(4). If results of a study cannot be reproduced by other independent researchers, the study has questionable validity.

Q373 Safety sampling is a technique using statistics to measure:

1. Accidental injury/illness performance
2. The effectiveness of line manager safety activities
3. The risk potential for accidental injury/illness
4. Both 1 and 3

Ans(2). Safety sampling measures the effectiveness of line manager safety activities - but not in terms of accidents. It measures effectiveness by conducting periodic samplings of how safely the employees are working. Safety sampling is based on the quality control principle of random sampling inspection. The degree of accuracy is dictated by the number of samples taken.

Q374 In probability, if order is important, a permutation is used. What is the formula for a permutation?

1. $P = N! / (N - X)!$
2. $P = N! / [X! (N - X)!]$
3. $P = X! / (X - N)!$
4. $P = X! / [N! (X - N)!]$

Ans(1). The formula for a permutation is:
$P = N! / (N - X)!$

Q375 In probability, if order is not important, a combination is used. What is the formula for a combination?

1. C = N! / (N - X)!
2. C = N! / [X! (N - X)!]
3. C = X! / (X - N)!
4. C = X! / [N! (X - N)!]

Ans(2). The formula for a combination is:
C = N! / [X! (N - X)!]

Q376 Evaluate the following to predict which set of circumstances will produce severe injuries if they are not identified and controlled:

1. Routine work tasks
2. Unusual and/or non-routine work tasks
3. Supervised assembly operations
4. Mass-production machine shop

Ans(2).A number of recent studies have shown that severe injuries are fairly predictable in certain situations. One of these involves unusual, non-routine work. This includes the job that happens occasionally, or a one-of-a-kind situation. Non-routine work may arise in factory operations but normal controls that apply to routine work have little effect in the non-routine situation.

Q377 ALARA is most often applied in:

1. Industrial hygiene
2. Safety engineering
3. Health physics
4. Loss control

Ans(3). ALARA is the main tenet of health physics. ALARA stands for As Low As Reasonably Achievable.

Q378 PEL's are:

1. Pollutant Emission Levels
2. Established by NIOSH
3. Permissible Exposure Limits
4. Not used in industrial hygiene

Ans(3).PEL's are permissible exposure limits and are established by OSHA.

Q379 A study collected 347 death certificates of former workers. The study found that 22 workers died from a form of cancer where 9.7 would be expected. The remaining workers died of other causes where 337.3 would be expected. Using the chi-square statistic, determine the p level at which this finding is significant.

1. The finding is not significant at any level
2. The finding is significant at p less than 0.01
3. The finding is significant at p greater than 0.99
4. The finding is significant at p less than 0.05

Ans(2).$X^2 = \Sigma$ [(observed-expected)^2 / expected]
= (22 - 9.7)^2 / 9.7 + (325 - 337.3)^2 / 337.3
= 16.05

Referring to the Upper Percentage Points of the X^2 Distribution Table with one degree of freedom, 16.046 corresponds to a "p" much smaller than 0.01.

This means it is very unlikely that the finding was the result of chance and it is thus significant at a "p" less than 0.01.

Q380 When using the chi-square statistic, why would one use that portion of the table with values for one degree of freedom?

1. It is the most conservative approach when evaluating cancer risks
2. Only one of the expected numbers can be determined independently of the total number of outcomes
3. Both 1 and 2
4. None of the above

Ans(2). Degrees of freedom is determined by the number of expected outcomes which can be determined independently of the total number of outcomes.

Q381 The statistical correlation coefficient is:

1. A constant used to correlate two dependent variables
2. The percentage of hood entry loss associated with different ventilation hoods
3. Used to develop a predictive model between two variables
4. An index of the extent to which two variables are associated

Ans(4). The correlation coefficient is an index of the extent to which two variables are associated.

Q382 Which of the following is NOT true about the correlation coefficient?

1. It can range from -1.0 to +1.0
2. Positive means a direct correlation
3. Negative means no correlation
4. As the correlation approaches 1.0, the association is stronger

Ans(3). A correlation of 0 (zero) means there is no correlation, (i.e., the variables show complete independence from one another). A negative correlation means the variables are inversely related to one another (e.g., as one increases, the other decreases).

Q383 A correlation coefficient of 0.95 means two variables are:

1. Weakly and directly correlated
2. Strongly and directly correlated
3. Weakly correlated in an inverse manner
4. Strongly correlated in an inverse manner

Ans(2). The two variables are strongly correlated because the coefficient of correlation almost approaches 1.0.

Q384 A regression equation is used to:

1. Determine the slope of a straight line
2. Determine the degree of correlation between two variables
3. Develop a predictive model between two variables
4. Determine the extent of deviation

Ans(3).A regression equation is used to develop a predictive model between two variables.

Q385 The z-test is used:

1. Only with normal distributions
2. To measure the distance of a particular value from the mean
3. Both 1 and 2
4. None of the above

Ans(3).A z-test is used on normal distributions to measure the distance of a value from the mean.

Q386 The t-test is used:

1. Only with normal distributions
2. To test the hypothesis that the mean score from a group does not differ from the expected mean
3. Both 1 and 2
4. None of the above

Ans(3).A t-test is used with normal distributions to test the hypothesis that the mean score does not differ from the mean. It is used to determine statistical significance.

Q387 In a group of 1,000 adults, 630 are women. What is the probability that a person chosen at random from the group will be a man?

1. 0.185
2. 0.370
3. 0.630
4. 0.740

Ans(2). If 630 people out of the 1,000 are women then 1,000 minus 630 (370) are men. The probability of choosing a man is

P = 370 / 1,000 = 0.370

Q388 What is the statistical distribution that considers the probability that an event will occur a specified number of times OR not at all? It is used for events that occur randomly.

1. Standard distribution
2. t-Distribution
3. s-Distribution
4. Poisson distribution

Ans(4).The Poisson distribution is the statistical technique to address this event.

Q389 Given ignition and that a fire does not terminate (probability = 0.80) and automatic sprinklers do not extinguish the fire (probability = 0.05) and manual suppression does not extinguish the fire (probability = 0.60), what is the probability that the fire will reach full involvement?

1. 0.024
2. 0.45
3. 0.55
4. 1.45

Ans(1). Use the formula.

P = ABC = Pterminates * Psprinklers * Psuppression = 0.80 * 0.05 * 0.60 = 0.024

Q390 The National Institutes of Health (NIH) "Recombinant DNA Research Guidelines" requires the formation of what type of administrative body?

1. Biohazard Safety Committee
2. DNA Candidate Screening Committee
3. Recombinant Czar
4. OSHA Staff Officer Review

Ans(1). The NIH guidelines require the formation of a Biohazard Safety Committee. This Committee serves as a technical resource to investigators, develops biohazard safety manual, and reviews research proposals.

Q391 Who among the following is REQUIRED to be on a process hazard analysis (PHA) team specified in OSHA's process safety management standard (29 CFR 1910.119)?

A. At least one operations/maintenance line employee (hourly, non-supervisory)
B. At least one operations/maintenance supervisor
C. At least one operations/maintenance engineer
D. At least one safety representative/safety manager
E. At least one person knowledgeable in the PHA methodology used
F. At least one person knowledgeable of the process being analyzed
G. At least one person capable of representing the company from a legal perspective
H. At least one external regulator (OSHA compliance officer, fire marshal, etc.)

1. A, E, F
2. E, F
3. All listed are specifically required
4. None listed are specifically required

Ans(2).No corporate positions are specifically required by the standard; however, the standard requires a team, and the team must have, at a minimum, one person knowledgeable in the PHA methodology (i.e., what-if/checklist, HAZOP, FMEA, etc.) and one person knowledgeable of the process being analyzed. Good practice would suggest composing a PHA team with the people occupying some of the other positions in this list.
29 CFR 1910.119 (e)(4)

Q392 Contract employers have specific responsibilities under the OSHA Process Safety Management Standard. These responsibilities include all of the following EXCEPT:

1. Assuring employees are trained in work practices
2. Assuring employees are instructed in the Emergency Action Plan
3. Conducting process hazard analyses of contract work
4. Advising facility employees of unique hazards discovered3

Ans(3). The contract employer is not required to perform the process hazard analyses. It is the responsibility of the employer that owns the applicable processes to conduct the process hazard analyses. The other answers are contract employer responsibilities.

Q393 Means of egress, as defined by the National Fire Protection Association (NFPA) Life Safety Code (NFPA 101) consists of three separate and distinct parts. Which of the following should be excluded?

1. Exit access
2. The exit
3. The exit discharge
4. The critical exit

Ans(4). Means of egress is a continuous and unobstructed way of exit travel from any point in a building or structure to a public way and consists of three separate and distinct parts, namely, exit access, the exit, and exit discharge.

Q394 Section 5 (a)(1) of the Occupational Safety and Health (OSH) Act:

1. Gives the Occupational Safety and Health Administration (OSHA) the right of entry
2. Defines responsibility for research
3. Is commonly called the general duty clause
4. Both 1 and 3

Ans(3). Section 5 (a)(1) of the OSH Act is commonly called the general duty clause and specifies the requirement that employers keep their workplaces free from recognized hazards that are causing or likely to cause death or serious physical harm.

Q395 ISO-9000 compliance has now become very popular in the United States and in other countries. What do the standards associated with ISO-9000 address?

1. company's methods for environmental management
2. A company's methods for system quality management
3. A company's methods for system safety management
4. All of the above

Ans(2). ISO-9000 standards are designed to enhance a company's quality performance by documenting all aspects of the company's operations. Though safety and environmental

performance enhancements are possible, the focus of ISO-9000 is to enhance quality by defining the processes and management systems necessary to produce a company's product. ISO-14000 standards deal with environmental management methods. Other ISO standards dealing with various aspects of safety management also exist and are expected to become more popular in the United States in the future.

Q396 After establishing the required management systems, to become formally registered as compliant with ISO-9000 or ISO-14000 standards, a company must:

1. Certify in writing to the International Organization for Standardization (ISO) that the company complies with the relevant standards.
2. Hire an ISO-certified third-party registrar to perform an external audit of the company's facilities at which point the third-party registrar will certify to the ISO that the company complies with the relevant standards (if the company is compliant).
3. Hire an ISO-trained auditor to perform an external audit of the company's facilities at which point the external auditor will certify that the company complies with the relevant standards (if the company is compliant).
4. Simply maintain compliance with the relevant ISO standards and submit an annual certification fee to ISO.

Ans(2). A registrar formally certified by ISO as a registrar must be retained for the audit and subsequent certification to ISO that the company meets ISO requirements.

Q397 By legislative fiat, who has the responsibility for conducting educational programs to ensure an adequate supply of qualified safety and health professionals?

1. NIOSH
2. OSHA
3. Private sector organizations
4. No agency has been designated

Ans(1). NIOSH, in the Department of Health and Human Services, has in its charter the directive to train safety and health professionals and to ensure that an adequate supply of these trained professionals are available.

Q398 By legislative fiat, who trains employers and employees in matters of safety and health?

1. NIOSH
2. OSHA
3. Trade organizations
4. Private sector

Ans(2). This duty is described in the OSH Act at Section 21 (c).

Q399 In the area of automobile safety, which agency is responsible for reducing accidents on the nation's highways?

1. DOT
2. NHTSA (National Highway Traffic Safety Administration)
3. EPA
4. OSHA

Ans(2). The NHTSA was established in 1970 to oversee the nation's highway safety program.

Q400 NIOSH is located in this federal agency:

1. Department of Health and Human Services/CDC
2. Department of Labor/OSHA
3. Department of Health and Human Services/PHS
4. EPA

Ans(1). NIOSH is located within the Department of Health and Human Services, Center for Disease Control.

Q401 OSHA approves:

1. Respirators
2. Safety-toed footwear
3. Safety hard hats
4. OSHA does not approve personal protective equipment

Ans(4). OSHA does not approve any personal protective equipment. NIOSH approves respirators, and several ANSI standards address the design requirements for the listed personal protective equipment.

Q402 OSHA standards are promulgated by:

1. OSHRC
2. NIOSH
3. EPA
4. Secretary of Labor

Ans(4). The Secretary of Labor promulgates OSHA standards.

Q403 The Occupational Safety and Health Administration (OSHA) is a part of the _____.

1. Department of Labor
2. National Institute for Occupational Safety and Health (NIOSH)
3. Department of Health and Human Services
4. Public Health Service

Ans(1).OSHA is located within the Department of Labor.

Q404 The Public Health Service (PHS) plays the following important role in occupational safety and health:

1. Many OSHA inspectors are in the PHS
2. Many NIOSH employees are in the PHS
3. AIDS research
4. The PHS has prime occupational safety and health responsibility

Ans(2). Many NIOSH employees are also commissioned officers in the PHS.

Q405 What agency governs man-made radionuclide use?

1. Environmental Protection Agency (EPA)
2. Nuclear Regulatory Commission (NRC)
3. Occupational Safety and Health Administration (OSHA)
4. Department of Energy (DOE)

Ans(2). The NRC governs radionuclide use if the materials are man-made. If the material is a naturally occurring radionuclide, then OSHA has jurisdiction.

406. As established by the OSH Act of 1970, the Occupational Safety and Health Review Commission is part of

1) The Executive Branch
2) NIOSH
3) OSHA
4) The Department of Labor Solicitor's Office

Ans.1 As part of the separation of power, this commission is part of the Executive Branch of the federal government.

407. A manufacturer of lawn, leaf, and brush shredders has just learned that an interlock on the shredder guard is defective and may not shut down the unit when the guard is removed or unintentionally works loose. The company is required by the CPSC to

1) Recall the shredders and fix them
2) Provide a replacement switch to product owners
3) Notify the CPSC (Consumer Product Safety Council) of the defect
4) Place ads in local papers alerting owners of the defect

Ans.3 While the other answers may be "good business", the Consumer Product Safety Act requires that the CPSC be notified by the manufacturer of such defects.

408. A listing of Nationally Recognized Testing Labs is published by

1) NIOSH
2) ANSI
3) EPA
4) OSHA

Ans.4 This listing, compiled by OSHA, contains a listing of labs that can test samples sent to them for contaminant,s according to the protocols established by the OSH Act or adopted by OSHA.

409. Most employers now allow their employees to access the Internet for work-related activities. Which of the following are available on the Internet?

1) CBT safety courses delivered by commercial firms
2) Codified regulations and laws maintained by their respective government agencies
3) Discussion groups dedicated to the practice of safety and risk management
4) All of the above

Ans.4 The Internet provides a huge repository of information that safety professionals can use to carry out their duties. This is just a small slice of the information available.

410. Among the most useful modern data storage technologies is a media called "CD-ROM." Each unit of this media can store nearly 640 megabytes of information in an optical format. The format is superior to magnetic media (such as floppy discs, hard drives, and tape) because of the following
    1) The CD-ROM is extremely durable and cannot be easily damaged physically
    2) The CD-ROM is never susceptible to data degradation
    3) The CD-ROM is not susceptible to data loss from magnetic fields
    4) One CD-ROM is more compact than an equally-sized hard drive

Ans.3CD-ROMs are not susceptible to data degradation or data loss from magnetic fields (such as those near monitors and speakers). CD-ROMs do experience data degradation over time, depending upon the quality of the CD-ROMs used. CD-ROMs are quite delicate and can easily be scratched or marked if not properly protected. CD-ROMs are actually quite bulky when compared to the storage capacity of modern multi-gigabyte hard drives.

411. For a state plan to be approved by OSHA, the state plan must have regulations to cover the safety and health of state and local employees
    1) Always true
    2) True, except where the state petitions OSHA for relief from this requirement
    3) False, except where OSHA specifically mandates this requirement due to the state's history with developing state plans
    4) Always false

Ans.1 For OSHA to approve a state plan, OSHA requires the state to cover its own employees and those employees of all of its political subdivisions to a level at least as effective as the OSHA regulations that cover all other employees. Some states have state plans approved that ONLY cover state and local employees while allowing OSHA to maintain its jurisdiction over all other employees in the state (currently, these states are Connecticut, New Jersey, and New York).

412. AIHA laboratory accreditation is good for how many years?
    1) 1 year
    2) 2 years
    3) 3 years
    4) 5 years

Ans.2 The AIHA Industrial Hygiene Laboratory Accreditation process follows ISO 17025; Requirements for the Competence of Testing and Calibration Laboratories. Every two years, a multiple day site inspector audits both management and technical requirements for evidence and documentation of laboratory best practices. If testing and calibration laboratories comply with the requirements of this International Standard, they are deemed to operate a quality management system to their testing and calibration activities that also meets the principles of ISO 9001.

413. The Accident Prevention Manual is published by the following organization
   1) OSHA
   2) NIOSH
   3) NSC
   4) NFPA

Ans.2 The Accident Prevention Manual is published by the National Safety Council.

414. The precursor of the National Safety Council was the.
   1) American Society of Safety Engineers
   2) Association of Iron and Steel Electrical Engineers
   3) American Industrial Hygiene Association
   4) American Safety Council

Ans.2 The Association of Iron and Steel Electrical Engineers evolved into the National Safety Council.

415. Human errors in a system can be detected using the following system safety technique
   1) Failure modes and effects analysis
   2) Preliminary hazard analysis
   3) Fault tree analysis
   4) Critical incident technique

Ans.4 In using the critical incident technique, those involved in accidents or near misses are interviewed. A survey is made of those having similar experiences, and everyone involved is asked to describe near misses or mishaps they can recall.

416. The procedure in which a job is broken into its component parts, and each task is analyzed to determine if there is a better, safer way to accomplish the job, is called a
   1) Failure modes and effects analysis
   2) Job safety analysis
   3) Fault tree analysis
   4) THERP

Ans.2A job safety analysis (JSA) is a procedure used by safety professionals and supervisors to review job methods and uncover hazards that had either been overlooked or resulted from changes in operations and/or personnel. The procedure evolved from military applications in World War II.

417. The analysis technique where individuals are interviewed about accidents, near misses, and hazardous conditions is known as a(n)
   1) Event tree analysis
   2) Fault tree analysis
   3) Critical incident technique
   4) System safety analysis

Ans.3 Critical incident technique involves the examination of accidents and near misses to find hazards.

418. Electrical fires are one of the most common types of building fires in the United States. What is the most common type of electrical equipment involved in electrical fires?
   1) Heating equipment
   2) Cooking equipment
   3) Electrical distribution equipment
   4) Air conditioning and refrigeration equipment

Ans.3 Distribution equipment is the primary cause of most electrical fires since it is often hidden and it is usually energized constantly.

419. What is the most appropriate coding method when identifying 15 items, from a human engineering view?
   1) Color coding
   2) Geometric shapes
   3) Stereoscopic depth
   4) Standard color coding chart

Ans.2 Geometric shapes is the most appropriate coding method when identifying 15 items from a human engineering view.

420. When designing a task that will be repeated throughout a shift, it should be designed so that it does not consume more than what percent of a persons maximum physical capacity?
   1) 25%
   2) 40%
   3) 50%
   4) 70%

Ans.2 40% capacity is the acceptable limit on the design of a task that is long-term in nature.

421. Human factors engineers work to change tasks to improve efficiency. When applying these principles to reduce accidents or injuries, the hardest item to change through HFE principals is
    1) Displays
    2) Controls
    3) Human parameters
    4) Task functions and order

Ans.3 Humans are the hardest part of the system to change. Therefore, the engineer attempts to improve efficiency or safety by working on the other parameters or sub-systems.

422. The procurement/acquisition process is frequently overlooked as a stakeholder in the design of systems. At which point in the design process should representatives from procurement be integrated into the design team?
    1) At the very beginning
    2) When the system's design intents are well understood
    3) When the system's component specifications are established
    4) When the system is ready to be released for bid or fulfillment

Ans.2 Though bringing in representatives from procurement at the very beginning may provide some design insight, the best time to integrate procurement into the design process is when the actual design intent of the system is well understood. At this point, procurement can provide insight on specific component specifications and vendors that can provide the required components and services.

423. Shortly following a gas manifold cleaning operation in which alcohol was used, an explosion, followed by fire occurred. A contributing factor could probably be traced to the presence of which of the following materials (besides the alcohol)?
    1) Hydrogen
    2) Oxygen
    3) Nitrogen
    4) Phosgene

Ans.2 Oxygen is the oxidizer and it participates in the combustion with the alcohol.

424. System safety and occupational safety considerations are frequently not considered early in a system's design process. Why is this?

    A. Some design engineers and project managers do not consider system safety and occupational safety issues as design engineering components (i.e., design intent and how to achieve design intent do not require input from system safety and occupational safety professionals).
    B. As a result of corporate structures, some system safety and occupational safety professionals are not concerned about business outcome and are instead focused only on product safety.
    C. Some design engineers and project managers believe that early consideration of system safety and occupational safety considerations will result in increased system costs and schedule and are best addressed once the functional design is completed.
    D. Some design engineers and project managers believe that system safety and

occupational safety considerations can be best addresses only after the system is in operation

1) C and D
2) A, C, and D
3) B
4) A

Ans.4 Unfortunately, many project managers and design engineers have no formal training in system safety and occupational safety and are unaware of the value (and ultimate savings) that can be realized by integrating such considerations early in the project life cycle. Early safety consideration can prevent future costly design changes.

425. A primary reason for conducting an accident investigation is to
1) Reduce the cost for workers' compensation
2) Find and fix the blame
3) Identify the cause and take actions to prevent recurrence
4) Prepare reports needed for litigation

Ans.3 Although there are a variety of accident investigations and analysis techniques, they all serve to identify root cause (or multiple root causes) and identify countermeasures that will prevent accident recurrence.

426. The LEAST likely root cause of accidents generally is a result of
1) Management system failures
2) Environmental factors
3) Physical limitations
4) Lack of a safety policy or program

Ans.4 Accident causation is very diverse, but generally each accident is a result of a combination of factors. The lack of a safety program, per se, is not a root cause. The management system allowing no safety program, though, is a primary root cause of accidents.

427. Safety professionals use accident investigations to
1) Prepare for OSHA inspections
2) Collect accident performance data
3) Identify the hazards that caused the accident
4) Prepare defense for litigation

Ans.3 Accident investigations are used to determine the root cause(s) of an accident, so that issues can be corrected. Although the information in the other answer selections obtained during accident investigations may be useful, these are not the primary reasons for conducting investigations.

428. In business terminology, what is a tall organization?
1) A company with well-developed management systems
2) A company with many levels of management
3) A company with broad market diversification

4) A company with extensive vertical market mobility

Ans.2 A tall organization has many levels of management and a narrow span of control at each level. In the United States, the tall organization is being displaced by flatter organizational structures.

429. What is a DISADVANTAGE of having an external group evaluate the company's safety program?
1) More objective
2) Fresh perspective
3) Image as "critics "
4) Non-prejudicial

Ans.3 A real problem with having external groups evaluate the safety program is that they may be perceived as "critics" rather than as teammates.

430. What is a DISADVANTAGE of having an internal group evaluate the company's safety program?
1) Evaluation staff knows plant workers
2) Less obtrusive audit
3) Potentially less objective
4) Harder to work with

Ans.3 One main drawback of an internal safety program audit group is that they are less likely to be totally objective due to personal friendships and "closeness" to issues.

431. A pattern of behavior associated with a distinctive social position is commonly referred to as a:
1) Status
2) Role
3) Class
4) Caste

Ans.2 Each society has many social categories which are referred to as roles, of which, each is associated with a particular behavioral pattern.

432. The theory on human motivation which declares that workers are uninterested and unmotivated toward work, and to motivate them to work, a system of rewards and punishment is necessary, applies to:
1) McGregor's Theory Y
2) McGregor's Theory X
3) Herzberg's Job-Enrichment Theory
4) Likert's Theory

Ans.2 In an attempt to analyze how management personnel view human motivation, McGregor evolved the notion that there are two basic ways in which management can view the worker: Theory X assumes the worker is essentially uninterested and unmotivated to work. Theory Y assumes the worker is basically interested and motivated to work.

433. In occupational safety, _____ are particularly important from the point of view of hazard identification and recognition through various warning systems:
    1) Anthropometric considerations
    2) Visual displays
    3) Population stereotypes
    4) Codings

Ans.3 In the man-machine system, man functions as a controller. For many situations, a generalized response is given, like expecting a light switch to be turned on with an upward movement and off with a downward movement, or turning a dial clockwise to obtain an increase. Such responses are called population stereotypes, a behavioral response common to nearly everyone in a certain population.

434. Maslow's Hierarchy of Needs: Biological - security - social - ego/status - self-realization. Once a lower need is met, a person seeks the next higher need. All of the following factors complicate hierarchical motivation EXCEPT:
    1) Multiple Motivation - People are often motivated by more than one need, for example, seeking recognition from fellow workers and supervisor
    2) Psychosocial Needs - These are continuing needs such as recognition and affection
    3) Conflicting Motives - Some motives can cause conflict, for example, desiring a promotion but fearing responsibilities
    4) Withdrawal Needs - Removes self from the situation, physically or psychologically

Ans.4 Withdrawal needs is not one of the factors complicating hierarchical motivation.

435. Which of the following technical departments or functions normally perform an FMEA?
    1) Safety Engineering
    2) Reliability Engineering
    3) Quality Assurance Engineering
    4) Production Engineering

Ans.2 Reliability engineers are often tasked to perform a failure modes and effects analysis (FMEA).

436. The primary purpose of a safety inspection is to:
    1) Determine if compliance with governmental codes and regulations are being met
    2) Find out if safety rules are being obeyed
    3) Look for unsafe conditions and practices

4) Discover and correct hazarde

Ans.4 The primary purpose of a safety inspection is to locate hazards and quickly follow up with actions to correct them.

437. Of the following, the least likely reason to perform a health and safety audit is to:

1) Determine if codes, laws, and regulations are being complied with
2) Assess the over-all program and its conformance to accepted safety and health practices
3) Determine if company policies, procedures, and standards are being met
4) Make sure the company is prepared for an OSHA inspection

Ans.4 The purpose of an effective health and safety audit is to conduct a systematic, objective analysis of a company's health and safety practices that is directed toward well defined goals, namely: regulatory compliance, accident/injury/illness risk control, program activities, records analysis and company policies. Following the audit, a report is prepared in which a written assessment is made along with recommendations to improve health and safety management of any significant risks that were identified.

438. Environmental health and safety audits are a means to:

1) Identify weak performers in plant manager roles
2) Ensure legal and corporate program compliance
3) Advise top management of compliance status
4) Both 2 and 3

Ans.4 A well run audit program will ensure compliance with corporate and legal requirements and advise top management of the situation at the plant. The goal is not to identify weak plant managers but rather to identify strengths and weaknesses of the EH&S program and to assess the level of compliance. In addition, a compliance assurance plan is generated.

439. A safety program evaluation can be defined as:

1) An exercise designed to placate management
2) An audit to highlight deficiencies
3) A measure of the effectiveness of safety programs
4) A method to get rid of unsafe supervisors

Ans.3 A safety program evaluation is defined as a systematic method of measuring the effectiveness of a planned group of activities aimed at improving workplace safety. It is important to note that most program evaluations look at both the good and bad aspects of the program; in other words, a properly conducted safety program evaluation is a balanced evaluation.

440. The main reason for safety inspections is to:
1) Identify unsafe acts
2) Determine compliance with OSHA standards
3) Prepare for insurance audits
4) Identify and correct workplace hazards

Ans.4 The main purpose of safety inspections is to identify hazards and to correct them.

441. An in-depth method that may be used to evaluate a safety program is a:
1) Safety program audit
2) Safety inspection
3) Job-Task survey
4) Safety check

Ans.1 A safety audit is a comprehensive examination of the safety program. The audit should provide insight into the effectiveness of the program and identify where improvements can be made. The American Society of Safety Engineers (ASSE) publicizes a comprehensive safety audit process to evaluate and rate a safety program by considering Organization and Administration, Industrial Hazard Control, Fire Control and Industrial Hygiene, Supervisory Participation, Motivation and Training, Accident Investigation, and Statistics and Reporting Procedures.

442. The purpose of an upper limit switch on a hoist is to:
1) Shut down the hoist when the load is too heavy
2) Prevent the load block from being raised too high
3) Prevent the running rope from coming off the hoist drum
4) To ensure that the load is centered over the hoist

Ans.2 The purpose of an upper limit switch is to prevent the load block from being raised into the hoist mechanism (sheave). When the upper limit switch is near contact by the load block, the crane upward movement is automatically shut down in order to keep the load block from contacting the sheave.

443. In auditing plant operations, you find a machine operator being exposed to a high noise level (determined through sampling). What is the most expedient method that should be taken to protect the machine operator?
1) Stop the operation
2) Equip the operator with personal protective equipment
3) Engineer control measures
4) Limit the operator's time of exposure

Ans.2 For expedience and until such time as engineering or administrative controls can be implemented, personal protective equipment must be provided and used to reduce the noise to an acceptable level.

444. Occupational Health and Safety Management Systems standard 18001 is a produced by:
   1) ANSI
   2) ISO
   3) ASSE
   4) BSI

Ans.3 Occupational Health and Safety Management Systems standard 18001 is a produced by BSI. The British Standards Institute led the development of 18001 as a health and safety management system standard that is compatible with ISO 9001 and ISO 14001.

BS OHSAS 18001:2007

445. The British Standards Institute (BSI) developed and published the Occupational Health and Safety Management Systems standard 18001 because:
   1) Of 'customer' requests for a management standard that addresses occupational health and safety
   2) Of 'customer' concerns that the ISO management system standards were inadequate
   3) BSI is responsible for assuring that United Kingdom has business standards necessary for global competition
   4) ANSI/AIHA had developed an American standard (ANSI/AIHA Z10)

Ans.1 The British Standards Institute (BSI) developed and published the Occupational Health and Safety Management Systems standard 18001 because of 'customer' requests for a management standard that addresses occupational health and safety. BSI has stated its intention to withdraw the standard if an Internal Standard is published covering the subject matter. Occupational Health and Safety Management Systems standard 18001 is produced by BSI. The British Standards Institute led the development of 18001 as a health and safety management system standard that is compatible with ISO 9001 and ISO 14001.

BS OHSAS 18001:2007

446. The British Standards Institute (BSI) developed and published the Occupational Health and Safety Management Systems standard 18001 to provide for:
   1) Organizations to fill a gap, thus providing additional revenue streams to BSI, through the sales of the standard. This will be successful until ISO publishes a similar standard
   2) A structure for an organization to incorporate occupational safety and health into the organizational management process. The standard is designed to work in conjunction with ISO 14001 and ISO 9001

3) An organizational structure to incorporate occupational safety and health into the organizational management process. The standard was necessary because occupational safety and health program needs are not compatible with ISO 14001 and ISO 9001
4) A management structure for an organization to accomplish occupational safety and health, providing an independent verification of organizational safety programs that is not negatively influenced by the quality management system objectives

Ans.2 The British Standards Institute (BSI) developed and published the Occupational Health and Safety Management Systems standard 18001 to provide for a structure for an organization to incorporate occupational safety and health into the organizational management process. The standard is designed to work in conjunction with ISO 14001 and ISO 9001.
BS OHSAS 18001:2007

447. The Occupational Health and Safety Management Systems standard 18001 concept is based on the recognized quality concept of:
1) Verification and validation
2) Process documentation
3) Plan, do, check, act
4) Six sigma

Ans.3 The Occupational Health and Safety Management Systems standard 18001 concept is based on the recognized quality concept of plan, do, check, act.
BS OHSAS 18001:2007

448. The planning process of the Occupational Health and Safety Management Systems standard 18001 requires that:
1) An organization or company completes a planning activity on a recurring basis with the organization or company's board of directors
2) An organization assures daily planning of work activities to assure safe practices are reviewed and implemented
3) An organization or company to establish processes, implement and monitor all procedures and processes and take corrective action when problems occur
4) An organization establishes objectives and processes supporting the organizations' occupational health and safety policy

Ans.4 The planning process of the Occupational Health and Safety Management Systems standard 18001 requires that an organization establishes objectives and processes supporting the organizations' occupational health and safety policy.

BS OHSAS 18001:2007

449. The Occupational Health and Safety Management Systems standard 18001 is a management system standard that is directed at reducing risk through a process of continual improvement. The standard is not a regulation or law. As such:
1) Compliance with OH&S legal requirements is only a piece of an OH&S management system conforming to the Occupational Health and Safety Management Systems standard 18001

2) Compliance with OH&S legal requirements is only the key factor of an OH&S management system conforming to the Occupational Health and Safety Management Systems standard 18001
3) Compliance with OH&S legal requirements is irrelevant to an OH&S management system conforming to the Occupational Health and Safety Management Systems standard 18001
4) Compliance with OH&S legal requirements is generally part of, but not required for an OH&S management system conforming to the Occupational Health and Safety Management Systems standard 18001

Ans.1 The Occupational Health and Safety Management Systems standard 18001 is a management system standard that is directed at reducing risk through a process of continual improvement. The standard is not a regulation or law. As such, compliance with OH&S legal requirements is only a piece of an OH&S management system conforming to the Occupational Health and Safety Management Systems standard 18001. Section 4.3.2, of the standard requires that legal and other requirements be continually identified and taken into account when establishing processes, procedures and running an OH&S management system. BS OHSAS 18001:2007

450. The ANSI/ASSE Z490.1 standard:
1) Provides the additional depth necessary to integrate the ANSI/AIHA Z10-2005, American National Standard for Occupational Health and Safety Management Systems into quality and environmental management systems
2) Is a standard that covers the areas of accomplishing training, including design, development, delivery and evaluation of training
3) Is the American Society of Safety Engineers' alternative to the American Industrial Hygiene Association's standard on safety management systems
4) Provides a basis for training that assures compliance with laws and regulations

Ans.2 The ANSI/ASSE Z490.1 standard is a standard that covers the areas of accomplishing training including design, development, delivery and evaluation of training. ANSI/ASSE Z490.1-2001

451. ASSE is the:
1) Association of System Safety Engineers
2) Association for Safe and Secure Environments
3) American Society of Safety Engineers
4) American Safety and Security Engineers

Ans.3 ASSE is the American Society of Safety Engineers. ASSE is an organization serving its members and the public by providing information and action on occupational safety, health and environmental issues and practices. ASSE membership is made up of professionals that work to prevent workplace injuries and illnesses, to create safer work and leisure environments and to develop safer products to enhancing the quality of life.

452. The ANSI/ASSE Z490.1, American National Standard, Criteria for Accepted Practices in Safety, Health and Environmental Training should be applied to:
1) Organizational safety, health and environmental management organizations
2) Any training that has the potential to create a hazardous environmental for the students

3) Organizational safety, health and environmental training plans to assure compliance with laws and regulations
4) Safety, health and environmental training providers

Ans.4 The ANSI/ASSE Z490.1, American National Standard, Criteria for Accepted Practices in Safety, Health and Environmental Training should be applied to safety, health and environmental training providers. As with all ANSI standards, Z490.1 is voluntary and does not guarantee regulatory compliance.
ANSI/ASSE Z490.1-2001

453. The International Organization for Standardization is:
1) A worldwide federation of national standards bodies
2) An organization created by the European Union to advance the business needs of the EU
3) A federation of international government appointed representatives that develop consensus standards for application on a worldwide basis
4) A federation of government representatives that develop unilateral agreements for the application of business practices across international borders

Ans.1 The International Organization for Standardization is a worldwide federation of national standards bodies.
ISO WWW1

454. ISO 19011:2002, Guidelines for quality and/or environmental management systems auditing is designed for:
1) Use by organizations wanting to get independent certification of its management system. ISO 19011 is a requirement for quality management systems under ISO 9001 and recommended for environmental management systems under ISO 14001
2) Use as guidance for audit programs whether those programs are for internal or external audits
3) Use as guidance for audits conducted by third party certification organizations to assure valid management system certification
4) Use by regulators, providing a guide to the application of management system audits to compliance inspections

Ans.2 ISO 19011:2002, Guidelines for quality and/or environmental management systems auditing is designed for use as guidance for audit programs whether those programs are for internal or external audits.
ISO 19011:2002

455. The ISO 19011:2002, Guidelines for quality and/or environmental management systems auditing provides:
1) Guidelines that are statutory in nature and are not necessarily applicable in all circumstances
2) Guidelines that, when adopted, need to be applied as specifically designated in order to claim conformance with the standard
3) Guidelines that should be used differently based on the nature, size and complexity of an organization that is being audited

4) Guidelines to be applied when conducting combined quality and environmental management system audits

Ans.3 The ISO 19011:2002, Guidelines for quality and/or environmental management systems auditing provides guidelines that should be used differently based on the nature, size and complexity of an organization that is being audited. The guidelines may be applied to quality management system audits, environmental management system audits or combined management system audits. There is no specific requirement for the guidelines to be implemented in order to have a conforming management system.
ISO 19011:2002

456. ANSI/ISO/ASQ QE 19011S-2004, American National Standard, Guidelines for quality and/or environmental management systems auditing was developed by the ASQ to:
   1) Provide for the special needs of the U.S. business community in maximizing U.S. international competition
   2) Assure U.S. business competitiveness with the European Union
   3) Provide U.S. corporations with an auditing program that would accomplish regulatory compliance requirements, as well as management system audits
   4) Provide additional guidance to the International Standard for application to the full range of audit activities

Ans.4 ANSI/ISO/ASQ QE 19011S-2004, American National Standard, Guidelines for quality and/or environmental management systems auditing was developed by the ASQ to provide additional guidance to the International Standard for application to the full range of audit activities.
ANSI/ISO/ASQ QE 19011S-2004

457. Which of the following is not an indicator of the variability of observations in an experiment?
   1) Mean
   2) Standard deviation
   3) Range
   4) Upper and lower confidence limits

Ans.1 The mean is the arithmetic average of the observations; it tells you nothing of the spread of the data. The standard deviation is a measure of how the data points differ from the average. The range is the high and low limits of the observations, and the confidence interval incorporates a probability as well as a measure of variability.

458. Quality control has been labeled as the science and art of identifying and controlling variability. One measure of variability as used in this context is:
   1) The arithmetic mean
   2) The size of a lot
   3) The mode
   4) The variance

Ans.4 The variance is a measure of the average squared difference between the mean and the individual items in either a sample or a population. Squaring these distances gives weight to the variation in distance from the mean and eliminates the negative sign.

459. Which of the following measures of variability is not dependent on the exact value of every measurement?
1) Mean deviation
2) Variance
3) Range
4) Standard deviation

Ans.3 The range is the difference between the largest and smallest value in a group of data. It ignores the data between these values.

460. A major objective of safety analysis programs is to:
1) Determine cause from past accidents
2) Determine foreseeable accidents
3) Determine fault
4) Analyze all accidents including non-injury accidents

Ans.2 Safety analysis programs have as their chief objective, the determination of possible accidents from systems. They do not rely on past accidents to determine cause. Analyses do not determine fault. Analyses are performed on systems, not past accidents or incidents.

461. When designing a safety program for an organization, it is necessary to consider the psychological factors that influence success, such as motivation, emotion, individual differences, and attitudes.
1) True
2) False
3) Not enough information
4) Safety programs are not designed for organizations

Ans.1 To work effectively with people, it is necessary to understand them. Much of the success of a safety program depends upon its acceptance by those to whom it is directed. Conversely, program acceptance is dependent upon an understanding of psychological factors which influence program success. These have been identified as being: individual differences, motivation, emotion, attitudes, and learning processes.

462. One of the primary reasons for the loss of buildings to fire has been an inadequate chain of events for notifying the fire department. Which element of the chain is often at fault in this chain?
1) Detecting the fire
2) Deciding to inform the fire department
3) Sending the message

4) The correct receipt of the message by the fire department

Ans.3 Buildings have been lost because of insufficient attention to the method of notifying the fire department.

463. A manufacturing process can by laid out according to the product being produced or according to the process that is used to make several products. An advantage of process layout vs. product layout is:
   1) Accounting costs are easier to track
   2) Material handling costs are lower
   3) Hazardous processes can be isolated
   4) Product production scheduling is easier

Ans.2 These costs are lower because there is less need to transport materials during the process.

464. A job safety analysis is a technique to:
   1) Establish safety procedures for each job
   2) Determine physical demands of each job
   3) Identify hazards associated with each step of the job
   4) Improve the workplace environment

Ans.3 A job safety analysis is a procedure to make the job safer by identifying the hazards or potential injuries/illnesses associated with each step of the job, in addition to developing a solution for each hazard that will eliminate or control the exposure.

465. Approximately ___% of the MALE population has some form of color blindness.
   1) 1
   2) 2
   3) 4
   4) 8

Ans.4 About 8% of men and 1% of women are color deficient. Color deficiency is a difficulty distinguishing color hue, especially under dim light. Color blindness is the inability to discriminate between certain colors, usually red, green, and gray. Very few people (about 3 in 100,000) are actually totally color blind (see everything as one color in different intensities).

466. Common carriers (e.g., airlines, bus companies, and railroads) must exercise a high degree of care (care that a very prudent and cautious person would undertake for the safety of others). This principle is called:
   1) Slight care

2) Reasonable care
3) Care and great care
4) Exercise of due care

Ans.3 Care and great care is the principle behind a high degree of care: that a very prudent and cautious person would undertake for the safety of others. Common carriers must exercise a high degree of care.

467. A master is liable for any negligence of his servant acting within the scope of employment. This concept is called:
1) Master negligence
2) Strict liability
3) Master-servant relationship
4) Master-employee liability

Ans.3 Master-servant relationship is the concept that a master is liable for any negligence of his servant acting within the scope of employment.

468. Historically, under common law, an employer is obligated to provide employees with all BUT:
1) A safe place to work
2) Safe tools to work with
3) Semi-competent supervisors
4) Knowledge of any hazards

Ans.3 Under common law an employer has been obligated to provide employees with competent (not semi-competent) supervisors.

469. Under the doctrine of comparative negligence:
1) Damage is based on the party that was most at fault
2) Damage is awarded to the party that was least at fault
3) Damage is awarded only if a party is not at fault
4) Damage is apportioned according to the degree of fault of the various parties

Ans.4 Damages are awarded and apportioned to the degree of fault of each party in the law suit.

470. Failure to exercise a degree of care that an ordinarily careful and prudent person (reasonable man) would exercise is the definition of:
1) A tort
2) Negligence
3) Degree of care theory
4) Common law

Ans.2 This is the classic definition of negligence. There must be a legal duty to use care. A breach of that duty and injury or damage is negligence.

471. Manual materials handling injuries include all of the following EXCEPT:

1) Musculoskeletal disorders
2) Carpal tunnel syndrome
3) Crushing of hands and fingers
4) Dropping of objects on feet/toes

Ans.2 Carpal tunnel syndrome is a repetitive motion injury, not a materials handling injury. Materials handling injuries are from overexerting muscles, tendons, and ligaments, that result in musculoskeletal injuries or trauma from falling objects or being caught between objects.

472. With repeated exposure to noise of sufficient intensity, a permanent threshold shift (PTS) will gradually appear first at _____ Hz.
1) 200 Hz
2) 1,500 Hz
3) 2,000 Hz
4) 4,000 Hz

Ans.4 As the number of years of noise exposure increases, the hearing loss between 3,000 Hz and 6,000 Hz becomes more pronounced.

473. A temporary depression of the hearing is called a:
1) Noise Induced Hearing Reduction (NIHR)
2) Temporary Threshold Shift (TTS)
3) Permanent Threshold Shift (PTS)
4) Temporary Auditory Depression (TAD)

Ans.2 A temporary depression of the hearing is called a temporary threshold shift, or TTS.

474. Administrative work practice controls for heat stress include acclimatization. This requires exposing the worker to the work environment for how long?
1) Two hours per day for 1 to 2 weeks
2) Four hours per day for 1 week
3) One full week
4) Two full days

Ans.1 Experience has shown that workers do not tolerate a hot job very well initially, but develop tolerance rapidly, and full endurance by working in the heat for about two hours per day for a period of one to two weeks. A worker acclimatized to heat will have a lower heart rate, a lower body temperature, a higher sweat rate, and more dilute sweat.

475. One of the most helpful groups to determine potential product problems with an existing product is:
1) The sales force

2) The service department
3) The manufacturing department
4) The quality assurance department

Ans.2 The service department is often the first group to know about emerging problems, especially with new products.

476. The process of eliminating unsafe acts is referred to as:
1) Positive reinforcement
2) Feedback
3) Extinction
4) Discipline

Ans.3 Extinction is the process of eliminating a behavior, including unsafe acts.

Dan Petersen, Safety Management: A Human Approach

477. What kind of industrial hazard can lead to an accident resulting in either death OR an accident with no severe lasting effects?
1) Chemical
2) Mechanical
3) Electrical
4) Ionizing radiation

Ans.3 Depending on a number of conditions (e.g., skin thickness, surface moisture, part of body involved, duration of current flow through the victim, etc.), an electric shock can result in either death or no lasting effect except for body burns that generally heal.

478. The most critical step in any safety program evaluation is:
1) The development of the evaluation protocol checklist
2) Spending "quality" time with line employees
3) Walking-through the survey with the responsible supervisor before formally conducting the evaluation
4) Appointing a competent evaluation team leader

Ans.1 A safety program evaluation will fail or succeed based on the quality of the evaluation protocol/checklist. The protocol determines the areas to be examined, the level of detail, and the questions to be asked.

479. A common injury to the wrist involving the median nerve which produces tingling and numbness is:
   1) Wrist drop
   2) Carpal tunnel syndrome
   3) Raynaud's syndrome
   4) White finger

Ans.2 Carpal tunnel syndrome is an injury/inflammation of the carpal tunnel of the wrist. The median nerve is compressed resulting in numbness and tingling in areas served by the median nerve. Other symptoms are also common, i.e., pain and sometimes muscle atrophy.

480. A condition which results from exposure to welding fumes that is characterized by accumulation of metal oxides in the lung without apparent physical symptoms is called:
   1) Pulmonary atresia
   2) Siderosis
   3) Idiopathic pulmonary distress
   4) Byssinosis

Ans.2 Siderosis is a pigmentation of the lung due to iron oxide exposure. The pigmentation can be marked but there seems to be no physical detriment. However, some siderosis patients present with chronic bronchitis and dyspnea. Differential diagnosis is difficult.

481. Metal fume fever is a delayed type of disease with flu-like symptoms. It has been associated with exposure to all of the following EXCEPT:
   1) Magnesium oxide fumes
   2) Nascent zinc oxide fumes
   3) Zinc oxide fumes
   4) Thorium oxide fumes

Ans.4 Metal fume fever has been shown to be caused by exposure to freshly generated (nascent) fumes of zinc and magnesium and their oxides. Although zinc and magnesium are the most common causes, copper has been known to cause metal fume fever also.

482. As one ages there is a vascular and neural degeneration of the inner ear. This results in a decrease in hearing ability. This condition is called:
   1) Temporary Threshold Shift (TTS)
   2) Permanent Threshold Shift (PTS)
   3) Presbycusis
   4) Auditory Threshold Reduction (ATR)

Ans.3 Presbycusis is the decrease in hearing that is associated with aging.

483. To assess your company's loss control performance, several statistics are captured and analyzed. One of these is "Number of lost time cases experienced during the past year." This statistic is an example of a(n):
   1) Leading indicator
   2) Lagging indicator
   3) Optional indicator
   4) Occupational indicator

Ans.2 Lagging indicators (such as historical statistics) provide data on how well the loss control system has performed and is useful for understanding how well a management system change affected loss control performance. However, lagging indicators are not as useful for predicting future loss control performance.

484. Which of the following ergonomically related orthopedic problems leads to the greatest cost?
   1) Carpal tunnel syndrome
   2) Lower back syndrome
   3) Ganglion formation
   4) Trigger finger

Ans.2 Statistics indicate that low back pain is responsible for more lost time than any other ergonomically related problem. About one third of all disabling injuries are due to manual handling of objects.

485. In injury epidemiology, the term incidence rate is often used. Incidence rate is the number of _____ that occurred in a given period of time divided by the sum of the exposed time.
   1) New cases
   2) Prevalent cases
   3) All cases
   4) Specific cases

Ans.1 Incidence rate is the rate of development of new cases of a specific disease over a given time period. The denominator is normally person-hours worked or some other similar measure such as person-years, etc.

486. There are a number of management standards published by ISO. ISO is an acronym for:
   1) International Organization for Standardization
   2) International Standards Organization
   3) Indentifying Standardization Organization
   4) Organization de Standards International

Ans.1 ISO is an acronym for the International Organization for Standardization. ISO is a non-governmental organization.
ISO World Wide Web

487. The ISO Quality Management Standards are found in which series of standards?
1) The 6000 series
2) The 9000 series
3) The 14000 series
4) The 18000 series

Ans.2 The ISO Quality Management Standards are found in the ISO 9000 series. ISO 9001, Quality Management Systems - Requirements is the best known. The ISO 14000 series covers Environmental Management. 18000 series is not in the ISO standards. OHSAS 18001 and 18002 are Occupational Safety and Health Management Standards generally considered to have originated from England.
ISO World Wide Web

488. An EMS, referred to in ISO 14004, is a system for:
1) Emergency Medical
2) Energy Management
3) Environmental Management
4) Emergency Management

Ans.3 ISO 14004 discusses guidelines for an Environmental Management System (EMS).
ISO World Wide Web

489. The ISO 9000 series of standards set forth and describe:
1) A regulatory framework that assures a corporation has a valid quality management system
2) A standardized list of expectations of high quality companies for use by governing bodies
3) A simple set of rules that can be followed by company officials to assure that employees accomplish quality work
4) A set of management practices recognized by international consensus

Ans.4 The ISO 9000 series of standards sets forth an internationally recognized series of standards that have been developed through a consensus process. The ISO 9000 series specifically identifies good quality management practices.
ISO World Wide Web

490. The ISO 9001:2008 standard provides:
1) A set of management system requirements that can be achieved and measured
2) A set of specific operating methodologies that companies must implement
3) A model organizational structure that will assure quality products and services
4) A method for auditing organizations to measure the organizational quality level

Ans.1 ISO 9001:2008 sets forth a system of requirements that an organization must achieve to be considered conformant with the standard. The standard does not specify how the organization must achieve the requirements or what the organizational structure must be.
ISO World Wide Web

491. The American National Standards Institute (ANSI) is:
1) The United States Government representative to international boards of inquiry and regulation
2) A United States non-profit organization
3) A United States industry advocate
4) A United States government organization that certifies standards

Ans.2 The American National Standards Institute (ANSI) is a 501(c)3 private, not-for-profit organization in the United States. ANSI oversees the creation, promulgation and use of standards and guidelines that impact every sector of businesses. ANSI is also involved in accreditation programs related to conformance to standards - including globally-recognized programs like the ISO 9000 (quality) and ISO 14000 (environmental) management systems.
ANSI World Wide Web

492. The American National Standards Institute (ANSI) oversees the development of national standards affecting businesses in the United States. ANSI's standards development process focuses on:
1) The selection of key experts for standards committees to assure scientifically sound standards are promulgated
2) The use of European Union approved methods to achieve global dominance of key industries
3) The use of consensus and broad based public involvement to achieve balanced and high quality standards
4) The use of standards to further the business interests of ANSI member companies

Ans.3 The American National Standards Institute (ANSI) oversees the development of standards affecting businesses in the United States. ANSI's standards development process focuses on the use of consensus and broad based public involvement to achieve balanced and high quality standards. The ANSI standards development process is designed to withstand scrutiny.
ANSI World Wide Web

493. The American National Standards Institute (ANSI):
1) Serves as the Head of ISO and chairs the IEC
2) Is not involved in international standards, as it is a United States based organization
3) Provides oversight for all ISO standards development
4) Is the US representative to ISO and IEC

Ans.4 The American National Standards Institute (ANSI) is the U.S. representative to ISO and IEC. ANSI is the U.S. representative of the two major non-treaty international standards organizations, the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC) [via the U.S. National Committee USNC)].
ANSI World Wide Web

494. AIHA is the:
1) American Industrial Hygiene Association
2) American International Hygiene Agency
3) Agency International for Hygiene and Assessment
4) Agency Industrial and Health Audits (from ISO)

Ans.1 AIHA is the American Industrial Hygiene Association. AIHA is an international, membership organization serving professionals that practice industrial hygiene. AIHA is the sponsor of the ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems.

AIHA World Wide Web

495. The ANSI Z10 standard is:
1) The American National Standard Criteria for Accepted Practices in Safety, Health, and Environmental Training
2) The American National Standard for Occupational Health and Safety Management Systems
3) The American National Standard version of the ISO 140001, Environmental Management Systems
4) The American National Standard equivalent of the ISO 9001, Quality Management Systems

Ans.2 The ANSI Z10 standard is the American National Standard for Occupational Health and Safety Management Systems. Z10 is the equivalent to the British Standards Institute, BS OHSAS 18001.
AIHA World Wide Web

496. The objective of the ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems is which of the following?
1) Provide a framework of auditing tools to measure the level of success of an organizations occupation safety and health programs
2) Set forth a strict set of practices and processes that must be implemented that will allow an organization to be certified as having an occupation safety and health management system
3) Provide organizations with a model to allow them to integrate occupational health and safety management into their overall business management system
4) Establish a framework for organizations to manage occupational health and safety irrespective of the success or failure of the business

Ans.3 The objective of the ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems is to provide organizations with a model to allow them to integrate occupational health and safety management into their overall business management system.
AIHA World Wide Web

497. The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems provides a system of:
   1) Detailed specifications to be implemented in each organization adopting the standard
   2) Strict auditing specifications for management system certifications
   3) A template for occupational safety and health consultants to use in establishing consulting businesses
   4) Principles with broad applicability to different organizations

Ans.4 The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems provides a system of principles with broad applicability to different organizations. Management system standards must be capable of being used by a wide variety of different organizations, as such, they are performance based, not prescriptive.

AIHA World Wide Web

498. ANSI, or the American National Standards Institute:
   1) Approves standards developed sponsors, granting approval once specific standards development criteria are verified
   2) Promulgates and publishes standards that are intended to be used by local governments in the development of regulations
   3) Promulgates and publishes standards that are developed in cooperation with various sponsoring organizations
   4) Works closely with the American National Standards Verification Board to assure the standards that it develops meet the specific criteria required of ANSI standards

Ans.1 ANSI, or the American National Standards Institute, approves standards developed sponsors, granting approval once specific standards development criteria are verified. ANSI does not develop standards and does not provide interpretations. Standards are developed under the direction of sponsors. Standards may be approved by ANSI only after ANSI has verified that the requirements for due process, consensus and other criteria have been met in the development process.
ANSI World Wide Web

499. The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems purpose is to:
   1) Establish performance criteria for management acceptance of risk to employees and the public
   2) Establish a tool for continual improvement of occupational safety and health performance
   3) Establish a tool to assist organizations with regulatory compliance
   4) Establish a minimum level of performance for organization health and safety programs in the international environment

Ans.2 The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems purpose is to establish a tool for continual improvement of

occupational safety and health performance.
ANSI/AIHA Z10-2005

500. The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems concept is based on the recognized quality concept of:
1) Verification and validation
2) Process documentation
3) Plan, do, check, act
4) Six sigma

Ans.3 The ANSI/AIHA Z10, American National Standard for Occupational Health and Safety Management Systems concept is based on the recognized quality concept of plan, do, check, act.
ANSI/AIHA Z10-2005

501. There are 10 plants in a manufacturing company. Four are to be picked at random to send a representative to safety planning meetings. How many different combinations of plant representation are possible at the meeting?
1) 210
2) 440
3) 720
4) 5,040

Ans.1 Because order is NOT important, this is a combination (instead of a permutation).

Combinations = N! / [X! (N-X)!]

(where N is the total number of plants and X is the number of plants chosen for representation at the safety planning meetings)

Combinations = 10! / [4! (10-4)!] = 3,628,800 / (24 * 720) = 210

502. In manual materials handling, methods for eliminating the need to bend include all of the following EXCEPT:
1) Using lift tables, work dispensers, and other mechanical aids
2) Raising the work level to an appropriate height
3) Providing materials and tools in front of the worker
4) Lowering the worker

Ans.3 Although providing materials and tools in front of the worker helps to prevent twisting motions, to avoid bending motions the work needs to be at the proper height. This can be accomplished through mechanical means (lift tables), raising the work level, or lowering the worker to the level of the work.

503. Human factors engineering or ergonomics is concerned with fitting the worker to a task or the task to the worker. A man/machine system is considered a closed loop system. Which one of the following does NOT make up the closed loop system?
   1) Reactions
   2) Data input
   3) Senses of man
   4) Size of material

Ans.4 The closed loop system is made up of reactions, machine, data input, and senses of man.

504. Human factors engineering is fitting the worker to the task or the task to the worker. Man is better at all of the following EXCEPT:
   1) Exerting large forces smoothly
   2) Handling the unexpected
   3) Sensitive to a wide variety of stimuli
   4) Flexibility

Ans.1 Machines are better at exerting large forces smoothly.

505. Human factors engineering is a technical discipline which concerns any of the following EXCEPT:
   1) Interrelationships between people
   2) Design of the workplace to reduce mistakes and accidents
   3) Design of tools to promote comfort
   4) Reduction of injuries

Ans.1 Human factors, also referred to as ergonomics, human engineering, bioengineering, and biomedics is a technical discipline mainly concerned with optimizing the relationship between people AND their work environment.

506. A safety program begins by:
   1) Developing a policy statement
   2) Assuring compliance with all OSHA standards
   3) Soliciting support of the facilities department
   4) Training all employees

Ans.1 Management must develop a policy statement that clearly defines management's support for safety and the safety-related roles of each member of the organization.

507. When an employee's exposure to an airborne, breathable chemical cannot be reduced to permissible safe levels through engineering controls, then the LEAST acceptable exposure control alternative is:
    1) Administrative controls
    2) Job rotation
    3) Changing production schedules
    4) Wearing a respirator

Ans.4 Administrative controls are methods for controlling employee exposures to contaminants or excess noise by limiting the duration of an individual's exposure by job rotation, work assignment, or time periods away from the contaminant or noise source. Personal protective equipment, such as a respirator, is the last alternative that should be considered to reduce exposures.

508. Safety directors should _____ as many levels of management as possible.
    1) Report to
    2) Have access to
    3) Have citation powers for
    4) Have budget approved by

Ans.2 The fact that the safety director's manager is interested in safety and supportive of the director is more important than the level he/she reports at. It is important that the company's upper management support safety and encourage safety. If the safety function reports at a high level in a company that does not value safety, the level of reporting will not matter. Access to managers throughout an organization will assist the safety director in developing an accepted, consistent, effective safety program.

509. Typically, the product safety engineer is a separate entity of which organization?
    1) Operations/Engineering
    2) Risk Management
    3) Quality Assurance
    4) Manufacturing

Ans.2 Generally the product safety engineer is a separate entity or of the risk management department organization. For a simple product or process, the product safety engineering

organization may be one person. If the organization is large or complex, several people may be needed, even with specializations.

510. The system safety responsibility in many company organizations is placed in the:
1) Engineering department
2) Product assurance department
3) Human factors engineering department
4) Manufacturing department

Ans.2 The system safety responsibility in many company organizations is placed in the product assurance department because it is an organizational method of establishing a close relationship among common disciplines such as reliability, maintainability, quality assurance, and human factors engineering.

511. Ozone has a propensity to form free radicals in the body. A toxicologic phrase utilized to describe this phenomena is:
1) Radiomimetic
2) Reactive
3) Chelator
4) Irritant

Ans.1 Ozone's propensity to form free radicals is termed radiomimetic. Webster's 3rd New International Dictionary defines radiomimetic as "producing, especially, biological effects similar to those produced by radiation."

512. Numerous LIFESTYLE factors have been shown to impact the toxicity of chemicals. For example, _____ is known to potentiate both the carcinogenic and noncarcinogenic effects of asbestos.
1) Smoking
2) Ethanol consumption
3) Coffee consumption
4) Obesity

Ans.1 Epidemiologic studies have shown that smoking dramatically increases the risk of developing fibrotic lung disease (asbestosis and bronchiogenic carcinoma). Mesothelioma, another tumor type associated with asbestos exposure, has an uncertain relationship to cigarette smoking.

513. Accidental contact with 60 Hz current of 5 to 25 mA can cause the following:
1) Death in most adults
2) Cardiac fibrillation
3) Paralysis
4) No effect

Ans.3 Victims may become paralyzed with accidental contact with 60 Hz of current in the range of 5 to 25 mA.

514. Electrical shock intensity and effects will depend on the:
1) Age of the conductor
2) Length of the conductor
3) Current path, frequency, and duration
4) Voltage drop across the conductor

Ans.3 Electrical shock intensity and effects depends on the current path, frequency, and duration.

515. Special hazard terminology has been developed for cranes and derricks. What is the correct term for an undetected hazard, created as a result of the crane's design or by the crane's use?
1) Dormant
2) Armed
3) Active
4) Either 2 or 3

Ans.1A dormant hazard is an undetected hazard created either by the crane's design or crane use. Armed and active are other types of crane and derrick hazards.

516. Which of the following are safety devices installed in boilers to relieve pressure?
    1) Blow down pipes
    2) Blow down valves
    3) Fusible plugs and safety valves
    4) Pressure relief pumps

Ans.3 Fusible plugs are designed to relieve pressure and to indicate certain conditions that contribute to low water (which causes pressure build-up). Safety valves will automatically relieve excess pressure or vacuum, that would otherwise damage equipment or result in injury to personnel. The selection, fabrication, installation, testing, and replacement of safety valves and fusible plugs should be in accordance with the ASME code.

517. Which is the preferred guard for power presses performing piercing, blanking, and stamping operations?
    1) Fixed
    2) Interlocked
    3) Adjustable
    4) Self-adjusting

Ans.1 The fixed guard is considered preferable to all other types and should be used in every case unless it has been definitely determined that this type is not at all practical. The fixed guard at all times prevents access to the dangerous parts of the machine. Besides power presses, fixed guards are also on guillotine cutters, drilling machines, gear trains, milling machines, and sheet flattening machines.

518. Guarding by means of a rail enclosure which restricts access to a machine is called:
    1) Point of operation protection
    2) Fixed guard
    3) Enclosure guarding
    4) Fencing

Ans.4 Guarding by means of a rail enclosure which restricts access to a machine is called fencing.

519. Which of the following statements describes ergonomics/HFE?

I. Engineering something for the population that will use it
II. Designing a system so that machines, human tasks, and the environment are compatible with the capabilities and limitations of people
III. Designing the system to fit the characteristics of people rather than retrofitting people into the system

1) I
2) II
3) III
4) I,II,and III

Ans.4 Ergonomics/HFE is a multidisciplinary activity dealing with interactions between man and his total working environment plus stresses related to such environmental elements as atmosphere, heat, light, and sound, as well as workplace tools and equipment.

520. Placing importance on individual differences, a work station or system should be designed to fit _____ % of the population:

1) 50
2) 90
3) 95
4) 99.7

Ans.2 When a system is designed which is directed at the average individual, much of the population will not be able to use the system efficiently because each person differs from the other; in other words, there is variability in the population. A better approach when designing a system is, instead of setting the standards to fit the average, design it to fit all but the upper 5% and the lower 5%. The system will then fit 90% of the population.

521. Anthropometry provides the safety engineer with data about:

1) Lifting capabilities of males and females
2) Dimensions of human body members
3) Historical data about accidents

4) Energy required to perform tasks

Ans.2 Anthropometric data provided specific data about the dimensions of the human body.

522. The most effective time to apply the principles of ergonomics is:
1) In accident investigations
2) During plant inspections
3) When planning a new operation
4) During special hazard evaluations

Ans.3 The most effective use of the principles of ergonomics is when planning a new operation or planning changes to an existing operation.

523. Ergonomics attempts to:
1) Redesign the product to be made more efficiently
2) Redesign tasks to fit the capabilities of males
3) Redesign tasks to fit the capabilities of humans
4) Redesign jobs to reduce the number of tasks required

Ans.3 The engineer attempts to redesign individual tasks in jobs so that a higher percentage of the human population can safely perform the tasks.

524. It is widely recognized that an important aspect of hand tool use is often overlooked. This aspect is:
1) Training
2) Tool repair
3) Tool design and selection
4) Tool labeling

Ans.1 Tenosynovitis is often called a "training" disease. Inexperienced or new workers often exert more force than necessary when using certain hand and power tools. Training in proper methods of tool use and encouraging workers to alternate hands when using selected hand tools (a design consideration, too) is encouraged.

525. What three concepts characterize ergonomics?
1) Human performance, ergonomic design, systems integration
2) Lifting, frequency, reach
3) Physical demands, mental demands, emotional demands
4) Academic, practitioner, integration

Ans.1 These concepts can be rephrased as work, engineering, and technology management.

526. Good engineering controls for manual materials handling tasks are based on which of the following principles?
1) Anthropometry
2) Epidemiology
3) Ergonomics
4) Physiology

Ans.3 Good engineering controls involve designing the workplace to fit the worker, one of the definitions for ergonomics. The key scientific areas that have been studied for the purpose of developing ergonomic guidelines for workplace design include: anthropometry, biomechanics, epidemiology, and physiology.

527. Work operations, reach, operator's position, and posture are all best analyzed by:
1) Civil engineers
2) Human factor engineers
3) Safety engineers
4) Environmental engineers

Ans.2 A human factors engineer would utilize these variables to conduct studies.

528. Which of the following is correct when isolating equipment through lockout/tagout?
1) Pulling a fuse on electrical equipment to guarantee a dead circuit
2) Disconnecting the main power source to fully disable most machines
3) Tightly shutting a high pressure valve before starting maintenance work
4) Physically locking the main power source and labeling it

Ans.4 The incorrect answers fail to adequately eliminate the hazard. Additional actions require physically "locking out" the energy source(s) and labeling it with a warning tag.

29 CFR 1910.147

529. How does a ground fault circuit interrupter work?

1) Senses small amount of current
2) Senses small amount of heat
3) Senses dampness
4) Senses unexpected resistance

Ans.1 The GFCI operates by sensing small amounts of current going to ground. There are various types of GFCI's with different tripping currents.

530. After a recent hazardous materials spill of hydrochloric acid, you have been asked to analyze the contents of a catch basin in which some of the spilled acid was collected. After collecting a 65 mL sample from the catch basin, you precisely separate the samples into two 30.0 mL pipettes. Next, you test the remaining 5 mL with PH paper and see that it clearly indicates a pH of around 1. You decide to perform an acid/base titration to determine the precise molarity, or molar concentration of the material in the catch basin. You prepare a 0.100 Molar solution of sodium hydroxide and titrate both of your 30.0 mL samples. The first sample requires 50.55 mL of sodium hydroxide for neutralization, and the second sample requires 50.45 mL. What is the molarity of the hydrochloric acid in the catch basin?

1) the pH paper indicated that the material in the catch basin is actually basic
2) 0.59 Molar
3) 0.17 Molar
4) 0.059 Molar

Ans.3 First of all, the indication of 2 on the pH paper implies an acidic liquid; therefore, an acid/base titration using sodium hydrochloric acid is appropriate. Sodium hydroxide and hydrochloric acid neutralize one another on a one-to-one basis (the one H^+ ion from HCI reacts with the one OH^- ion in NaOH, and the one Na^+ ion in NaOH reacts with the one CI^- ion in HCL). Because of this one-to-one relationship, we can determine the molarity by using this simple identity: $M_1V_1 = M_2V_2$. M_1 is 0.100 M of which 50.5 mL (average) was used to neutralize 30.0 mL (V_2) of acid of unknown molarity (M_2). $(M_1V_1)/V_2 = M_2$ (0.100 M * 50.5 mL)/30.0 mL = M_2 = 0.168 M

531. A design which allows the system to function safely until corrective action is possible is called:
   1) Fail-corrective
   2) Fail-passive
   3) Fail-operational
   4) Fail-non-operational

Ans.3 Fail-operational is a design which allows the system to function safely until corrective action is possible.

532. When derating an electronic part, which of the following techniques is not appropriate?
   1) Decreasing MTBF
   2) Derating for temperature
   3) Derating for stress
   4) Lower the power dissipation

Ans.1 Decreasing the MTBF is not an appropriate technique for derating an electronic part.

533. The scientific study of how, when, where, and why an illness distributes itself over time in a population describes:
   1) Biogenetics
   2) Epidemiology
   3) Etiology
   4) Health physics

Ans.2 Epidemiology is a science that investigates the distribution and causes of disease. Epidemiologists study the interplay between the physical environment, biological effects, and social phenomena that relate to human health and illness in the workplace.

534. A group of foundry workers are exposed to an air temperature of 105ºF. Which control would be least appropriate?
   1) Wear more clothing
   2) Decrease the air movement
   3) Increase air velocity
   4) Decrease the humidity

Ans.4 The principal heat load in a foundry is radiant. It is highly unlikely that foundry air will have much moisture. All other methods may have some net benefit.

535. TLV's for gases and vapors are estimated on what unit basis?
1) Weight to weight
2) Weight to volume
3) Volume to volume
4) Both 2 and 3

Ans.4 TLV's for gases and vapors are expressed in parts per million, which is a volume/volume ratio, and milligrams per cubic meter, which is a weight/volume ratio.

536. It is necessary for a maintenance worker to enter a confined space where a potential atmosphere immediately dangerous to life or health (IDLH) may be present. Which of the following respirators should the worker use?
1) NIOSH-approved full face chemical cartridge respirator with cartridges rated for the chemical hazard
2) NIOSH-approved full face demand SAR (Supplied Air Respirator)
3) NIOSH-approved full face pressure demand SCBA (Self Contained Breathing Apparatus)
4) NIOSH-approved half face cartridge respirator with HEPA cartridges

Ans.3 A pressure demand SCBA or pressure demand SAR is required in a known or suspected IDLH atmosphere. A demand SAR may sometimes allow contaminants to be sucked into the mask which may have a negative pressure with respect to the outside atmosphere. A pressure demand SCBA constantly supplies air to the face piece (to ensure that a negative pressure is never experienced inside the face piece). Cartridge respirators are not permitted in known (or suspected) IDLH atmospheres. 29 CFR 1910.134 (d)(2)

537. The basic safety precaution for people who work in confined spaces or in dangerous atmospheres is:
1) The buddy system
2) Use of SCBA's
3) Use of safety harnesses and life lines
4) Proper air sampling

Ans.1 The basic safety precaution for people who work in confined spaces or in dangerous atmospheres is the "buddy system". With the buddy system, no worker enters a dangerous atmosphere until someone else has been designated to take responsibility for his/her safety. While one worker goes into a confined space or other dangerous area, the buddy stays in a safe place nearby, alert at all times for danger and ready to act in an emergency. (Usually the

buddy is suited up in protective equipment, as is the worker, ready to help the worker out in case of trouble).

538. A mechanical filter respirator is designed to prevent entrance of _____ into the respiratory system.
1) Gases
2) Gases and dusts
3) Dusts and mists
4) Gases and fumes

Ans.3 Mechanical filters can only remove particles, they cannot remove gases or vapors.

539. Air-purifying and atmosphere-supplying are:
1) Types of respirators
2) Ventilation system designations
3) ASTM respirator classes
4) OSHA-approved respirators

Ans.1 Respirators can be placed in two categories: a) air-purifying, and b) atmosphere-supplying. Most air-purifying respirators use chemical cartridges to remove contaminants from the outside atmosphere to a concentration acceptable for breathing. Atmosphere-supplying respirators include SCBA and SAR. These respirators have an independent supply of breathable air for the respirator-wearer. There is no such thing as an OSHA-approved respirator.

540. Barrier creams:
1) Neutralize potential irritants
2) Reduce contact of potential irritants with the skin
3) Are preferred to gloves in most cases
4) Are not effective

Ans.2 Barrier creams, as their name implies, form a protective barrier between the skin and the chemical. They are not neutralizing agents for chemicals.

541. In which of the following situations may an air-purifying respirator be worn safely?

1) Odor threshold is greater than the allowable level
2) Odor threshold is equal to the allowable level
3) Odor threshold is less than the allowable level
4) Odor threshold is at least 4 times the allowable level

Ans.3 To wear an air purifying respirator safely, the worker must be able to detect the material at a safe level. If the odor threshold is greater than the allowable level (e.g., the odor threshold is 20 ppm, while the allowable level is 10 ppm), the worker will not know the material is getting through until PEL/TLV levels are reached or exceeded.

542. The risk of developing active pulmonary tuberculosis is substantially increased by which of the following?
1) Exposure to free crystalline silica dust
2) Exposure to alpha radiatioc
3) The presence of arsenin
4) Exposure to beta radiatioc

Ans.1 Exposure to free silica dust is a risk factor for pulmonary tuberculosis. The fibrosis caused by the silica may make the lung more susceptible to tuberculosis.

543. Which of the following would be most effective in reducing dermatitis in electroplaters?
1) Barrier creams
2) Gloves
3) Local exhaust ventilation
4) Frequent hand washing

Ans.2 The best method of protecting oneself from dermatitis is to prevent all contact with the offending agent. Therefore, gloves is the best answer.

544. When an internal explosion occurs, it causes the shape of the enclosure to change from:
1) Rectangular to elliptical
2) Rectangular to star-shaped
3) Rectangular to quadrilateral
4) Rectangular to pyramid

Ans.1 When an internal explosion occurs, it causes the shape of the enclosure to change from rectangular to elliptical. Adequate strength is one requirement for the design of an explosionproof enclosure; a safety factor of 4 is generally used. To prevent failure of the enclosure, openings are designed to relieve the pressure of the expanding gases.

545. You suspect that a worker's exposure has exceeded the TLV-C for a certain chemical. Unfortunately you do not have a device to measure instantaneous exposures. You do have a longer term measurement device. What do you do? Note: Substance does not cause an immediate reaction.
1) Take an 8-hr sample and back calculate
2) Take as short as sample as possible
3) Take a 15-min sample
4) Do not measure and assume the TLV-C was exceeded

Ans.3 If instantaneous monitoring is not feasible, then the TLV-C can be assessed by sampling over a 15-min period - except for those substances that may cause immediate irritation when exposures are short.
ACGIH TLVs and BEIs

546. Pressure testing is a potentially _____ test.
1) Destructive
2) Hypergolic
3) Hyperbaric
4) Reverberant

Ans.1 Pressure testing may actually destroy the object being tested.

547. How many footcandles of illumination are necessary for work that requires fine assembly?
1) 5 footcandle
2) 50 footcandle
3) 500 footcandle
4) 5,000 footcandle

Ans.3 500 footcandle is necessary for the performance of visual tasks of low contrast and very small size over a prolonged period of time (e.g., fine assembly, very difficult inspection, etc.). 5 footcandles is very low light, and is not appropriate for most working environments. 50 footcandle is a typical office environment lighting level, and 5,000 footcandle is well beyond the lighting necessary for the task described.

548. As a safety professional, you are asked to provide guidance in the layout of a 5,000 square-foot, high-hazard facility to be occupied by up to 15 laboratory scientists. The expected configuration of the process equipment presents the need for a dead end corridor in the means of egress arrangement. Based on NFPA 101 (Life Safety Code), what will be your suggestion regarding the maximum permitted length of the anticipated dead end corridor?
1) 0 ft
2) 25 ft
3) 30 ft
4) 50 ft

Ans.1 Dead ends are not permitted in high hazard occupancies with an area of more than 200 square feet or an occupant load of more than 3 people. NFPA 101 7.11.4 (Life Safety Code)

549. The ACGIH TLV-TWA Committee in 1996 and 1997 proposed which of the following as a change in the designation of carcinogenic chemicals?
1) No changes
2) No longer designate carcinogens
3) Established 5 categories of occupational carcinogens
4) No longer publish exposure limits for carcinogens

Ans. The Committee proposed that carcinogens be divided into 5 separate categories to better reflect degrees of uncertainty and scientific knowledge.
A1 - Confirmed Human Carcinogens
A2 - Suspected Human Carcinogens
A3 - Animal Carcinogens
A4 - Not classifiable as a Human Carcinogen
A5 - Not suspected as a Human Carcinogen

550. Delegation of authority and chain of command are characteristic of a:
1) Democratic structure
2) Formal structure
3) Autocratic organization
4) "Laissez faire" organization

Ans.2 The formal structure has a system of objectives that defines roles and responsibilities of the participants. Delegation of authority means authorizing individuals within the structure with defined rights and responsibilities resulting in a "chain of command".

551. All have been identified as "megatrends" by John Naisbett EXCEPT:
1) Transition from industrial to information society
2) National to world economy
3) Short term to long term
4) Decentralization to centralization

Ans.4 Naisbett predicted a number of changes including the movement from centralization to decentralization in government and business.

552. Heinrich's domino theory of accidents includes:
1) Worker, perception, physical, social, injury
2) The 5 B's
3) Social, fault, unsafe act, accident, injury
4) Men, machine, money, capital, accident

Ans.3 Heinrich's domino theory involved social, fault, unsafe act, accident, injury.

553. An effective fire loss control program must include objectives that prevent loss of life and personal injury, prevent inception of fire, protect property and:
1) Provide means to extinguish fire
2) Provide uninterrupted operations
3) Incorporate the use of fire resistive building materials
4) Effect controls that limit the fire load (amount of combustibles)

Ans.2 In any fire loss control program there is a necessity for a plan that protects against a business interruption. Generally, there is a condition of insurance that you reduce the loss resulting from a business interruption by resuming complete or partial operations on the fire damaged premises and/or making use of other property and materials at other facilities.

554. Political and socioeconomic factors are NOT important considerations in which of the following processes?
1) Risk communication
2) Risk assessment
3) Risk management
4) Risk acceptability

Ans.2 Risk assessment is the scientific discipline involved in characterizing and quantifying the magnitude of potential harm to human health from the presence of a particular agent in the environment. Unlike risk management, risk assessment does not consider economic impact.

555. What does the management term "line function" mean?
1) An employment function for which the enterprise exists
2) An employment function to support the work for which the enterprise exists
3) An employment function associated with a manufacturing assembly line
4) An employment function providing direct supervision

Ans.1 A line function is a position or role directly involved with the main enterprise of the company. For example, "pigment mixer" would be a line function in a paint manufacturing plant. A staff function, on the other hand, is a position or role that supports the line functions. For example, "security guard" is a staff function for a paint manufacturing plant to ensure the protection of the facilities and people involved in manufacturing paint.

556. A manager who utilizes an external reward and punishment system to affect performance is a _____ manager.
1) Theory A
2) Theory X
3) Theory Y
4) Theory Z

Ans.2 Theory X holds that people must be motivated to work by external reward and punishment.

557. A manager who utilizes the inherent motivation that is present in all humans is a _____ manager.
1) Theory A
2) Theory X
3) Theory Y
4) Theory Z

Ans.3 Theory Y managers utilize the goal-orientation of the worker to achieve results.

558. The tendency for an individual who finds himself in a frustrating circumstance and reverts to an earlier form of behavior - such as putting on a temper tantrum - is called:
1) Negativism
2) Aggression
3) Reactivity
4) Regression

Ans.4 Regression is the tendency for an individual who finds himself in a frustrating circumstance to revert to an earlier form of behavior.

559. The tendency for an individual to give up or withhold any sense of emotional or personal involvement in the situation is called:
1) Resignation
2) Aggression
3) Regression
4) Reactivity

Ans.1 Resignation is the tendency for an individual to give up, or withhold any sense of emotional or personal involvement in the situation.

560. Human performance is a function of:
1) Age and sek

2) Attitude and motivation
3) Training and motivation
4) (Aptitude + training) * motivation

Ans.4 Human performance is a function of (aptitude + training) * motivation.

561. It is possible to reduce incidents involving human error by:
1) Carefully screening job applicants
2) Furnishing workers with personal protective equipment
3) Providing training and education to workers and line supervisors
4) Improving workplace conditions

Ans.3 It is generally agreed that the majority of incidents leading to an accident are attributed to unsafe acts rather than unsafe conditions. Although general estimates of the mental and emotional states that cause accidents (like indifference, resentment of authority, disregard of danger, undue haste, anger, impatience, boredom, etc.) are not reliable, there is consensus that effective training and education will prevent many of the human errors (unsafe acts) that arise from mental and emotional states.

562. Workers' compensation represents a significant business expense for employers. To a large extent, employers can influence the cost of this coverage by:
1) Obtaining a low rate prospective experience rating plan
2) Improving their experience modification rate
3) Obtaining a low rate retrospective experience rating plan
4) Negotiating with life underwriters

Ans.2 One of the objectives of the workers' compensation system is to encourage safety in the workplace. A credit rating is awarded for better than average injury/illness accident prevention OR a debit rating is charged against the premium for one that is worse than average. Injury rates and the experience modification rating are used to estimate charges. By lowering the experience modification rate one can lower their insurance premium.

563. The acronym CAD/CAM has become very important to manufacturing. What does CAD stand for?
1) Cost and development
2) Computer aided design
3) Computer assisted development
4) Count all devices

Ans.2 CAD/CAM stands for computer aided design and computer aided manufacturing. CAD/CAM is used to make manufacturing more cost effective, more time efficient, and to increase overall product quality.

564. A safety professional is asked to evaluate a procedure to address a high pressure steam fitting leak. He or she should ensure that which of the following safety-related caution steps is included prior to repairs being made?
   1) Use an intrinsically safe wrench
   2) Ensure system is shut down, bled, locked out, and tagged out
   3) Wrap line with electrician's tape before making repairs
   4) Patch leak with non-hazardous epoxy

Ans.2 The correct caution step is to verify that the system is properly bled and to ensure that the lock out/tagout procedures have been implemented.

565. In organizing and implementing a successful safety and loss control program, it is essential that you:
   1) Obtain adequate funding
   2) Get top management support and participation
   3) Acquire an environmental health and safety staff
   4) Promulgate a formal safety policy and program

Ans.2 An effective safety program must begin at the top. Only top management can assign the responsibility for safety and loss control and ensure its performance. Also, top management can implement the controls in the workplace environment along with worker operations that will control accidents.

566. All of the following are examples of staff functions EXCEPT:
   1) Training
   2) Assisting
   3) Facilitating
   4) Commanding

Ans.4 Commanding is a line function and, therefore, not an example of a staff function.

567. A supervisor's job responsibilities include the "3-E's" principles. Which of the following would NOT be included under the "3-E's"?
   1) Making sure employees comply with safety rules and procedures
   2) Instructing employees how to work safely
   3) Establishing a safety program
   4) Determining ways for eliminating injury/illness potential from the job

Ans.3 The supervisor's job involves:

ENGINEERING - to determine the most effective way of doing the job and designing the injury/illness potential out of the job.

EDUCATION - to instruct the employee in the right method, and through training, improve job skill, knowledge, and behavior.

ENFORCEMENT - to ensure the employee follows the prescribed work practice and complies with safety rules and procedures.

568. Within a PERT network, slack time is defined as:
   1) Low production time
   2) Rest time
   3) The difference between the length of time scheduled for an event and the length of time required for completion
   4) The longest path between critical events

Ans.3 Slack time within the PERT network is defined as the difference between the length of time scheduled for an event and the length of time required to complete the event.

569. The critical path in a PERT network is defined as:
   1) The time slippage between tasks
   2) The start time between the longest tasks
   3) The longest time path between beginning and end events
   4) The shortest time path between beginning and end events

Ans.3 The critical path in a PERT network is the longest time path between beginning and ending events. Critical path defines the sequence of events that control the amount of time needed to complete the effort described by the PERT network.

570. The responsibility for IMPLEMENTING a company's safety and health policy is that of:

1) Top management
2) Line and operating management
3) Safety and health professionals
4) Both 1 and 2

Ans.2 Top management is responsible for setting the company's policy on safety. Top management sets clear-cut restrictions and regulations appropriate to the type of operation, processes, and company size to keep employees at all levels aware of what is required of them in regard to safety. Line and operating management are responsible for implementing this policy and are accountable to top management. Together they are accountable for the safety and health program. The safety professional is the technical expert and can provide advice and counsel on potential implementation plans, but does not generally implement them.

571. Responsibility for safety in day-to-day operations belongs to:

1) The assigned safety engineer
2) Top management
3) The line supervisor
4) Operations management

Ans.3 The line supervisor is the key person who plays an important role as the link between management and the worker. The worker follows the line supervisor's orders, and each worker is responsible for the quality and quantity of their own work.

572. It is generally accepted that the most effective safety program places safety responsibility on the:

1) Chief executive officer
2) Safety director
3) Supervisor
4) Employee

Ans.3 If a supervisor is held responsible for safety performance, he or she will have a personal interest in the program.

573. The success of the system safety effort depends on:
1) Employing applicable systems safety techniques
2) Implementing current design practices
3) Defining safety objectives
4) Identifying and correcting potential hazards

Ans.3 The success of the system safety effort depends on definitive statements of safety objectives and requirements by the managing activity and their translations into functional hardware and software.

574. Relatively new developments in human resources management include:
1) Flex-time
2) Job-sharing
3) Work-sharing
4) All of the above

Ans.4 New developments include flex-time (flexible working hours), job-sharing (two or more workers sharing the same job), and work sharing (during downtime, hours are taken away from everyone to prevent layoffs).

575. System engineers seek to drive down the probability of causing a death to users of a system. What is this probability?
1) 10^{-3}
2) 10^{-4}
3) 10^{-5}
4) 10^{-6}

Ans.4 To the average person, accidents with the probability of death of 10^{-6} or less per person per year are not of great concern to the average person. Hence, many systems safety experts are attempting to design products or processes to a level that is "ten to the minus six" level of performance, or one in a million.
NOTE: the probability of being killed in a motor vehicle accident hovers consistently around 10^{-4}.

576. The design, development, and delivery of a new product can be called:
1) System engineering
2) Manufacturing system
3) Production cycle

4) Designing for manufacturing

Ans.2 The design, development, and delivery of a new product can be called a manufacturing system. The system incorporates accidental loss consideration involving system integration of a Machine/Environment (occupational safety) with subsystems pre-production safety and quality reviews (product safety).

577. An excellent way to reduce hazardous material inventory in manufacturing is:
1) JIT
2) CAD
3) CAM
4) QA

Ans. JIT or Just-In-Time inventory control only delivers hazardous materials when such items are actually needed by the process, thus allowing for reduced hazardous materials storage requirements.

578. What is the most critical period in crisis management?
1) First 24 hours
2) First week
3) First month
4) First year

Ans.1 It is generally considered that the first 24 hours of any crisis are the most critical during the life of the crisis. Management's function is to explain the problem, control rumors, and control panic.

579. What is the value of an investment that will be worth $10,000 after 5 years, given that the interest rate during this period is 10%?
1) $3,209
2) $6,209
3) $12,000
4) $20,000

Ans.2 This question requires calculating the time value of money. Find the present worth of the future investment value as follows:

Use the formula: $P = F(1 + i)^{-n}$
$P = F(1 + i)^{-n}$
$= \$10,000\ (1 + 0.10)^{-5}$
$= \$6,209$

580. Organizations allocate costs into various categories to aid decision making. Often, these costs are determined as a percent or ratio of manufacturing costs. Examples include raw materials, direct labor, indirect materials, selling costs, distribution costs, and other costs. In calculating return on investment of ergonomic controls, a key expense used to determine the cost-benefit of controls is:

1) Direct costs of accidents inflated over time to determine present value of future workers' compensation injury dollars
2) The cost of capital associated with purchase and installation of controls
3) Depreciation costs and pretax operating expenses after controls are implemented
4) Profit and quality loss as a result of increased absenteeism and poor employee morale

Ans.2 While all are important, the cost of capital (interest lost from the initial investment) is often ignored. Knowledgeable financial managers will consider the cost of capital when calculating their desired return on investment of a capital expenditure. The challenge of the practitioner is to keep this in mind when determining cost-benefit of controls and, when possible, to quantify productivity efficiency and productivity enhancements to help support action plans.

581. At an annual interest rate of 6.0%, what must the total return on a $50,000 investment be over 3 years to break even?

1) $59,551
2) $56,180
3) $50,000
4) $53,000

Ans.1 This is a compound interest problem. If invested and left alone, the $50,000 would be worth $53,000 after year 1; $56,180 after year 2; and $59,551 at the end of year 3.
This is found as follows:

$50,000 * 1.06 = $53,000
$53,000 * 1.06 = $56,180
$56,180 * 1.06 = $59,551

582. It is estimated that the total cost to the country from an OSHA regulation is $20 million the first year and $5 million in each subsequent year. If the regulation is estimated to save 5 lives per year, what is the cost per life saved over a 10 year period, not considering the time value of money?

1) $1,000,000

2) $1,400,000
3) $1,300,000
4) One cannot put a value on human life

Ans.3 To get the cost per life, first calculate the total cost of the regulation.
The total cost of the regulation is $20 million for the first year plus (9 * $5 million) for each of the subsequent years. This is $65 million.
Then calculate the total lives saved. The total number of lives saved is (10 years * 5 lives per year) = 50 lives.
Then divide the total cost by the total lives saved.
Thus, the cost per life saved is $65 million / 50 lives = $1.3 million/life.

583. If an investment of $20,000 is made to buy ergonomic tools, and these tools should prevent $10,000 per year in injury-related costs, when will the investment pay for itself, assuming an annual interest rate of 10%?
1) At the end of 2 years
2) During the third year
3) During the fourth year
4) At the end of the first year

Ans.2 This question requires calculating the time value of money. Find the present worth of the future savings as follows:

Present worth of at the end of each year:
$PW = F (1 + i)^{-n}$
$PW(1) = \$10,000 (1 + 0.10)^{-1}$
$= \$9,090.90$
$PW(2) = \$10,000 (1 + 0.10)^{-2}$
$= \$8,264.46$
$PW(3) = 10,000 (1 + 0.10)^{-3}$
$= \$7,513.15$
$PW(1) + PW(2) = \$17,355.36$
$PW(1) + PW(2) + PW(3) = \$24,868.51$

The tools will pay for themselves at some time during the third year. Remember that PW(1) + PW(2) is the present worth at the END of Year 2.

584. A company's finance department requires that all investments pay for themselves within 24 months. You have recommended that $50,000 in material handling equipment be installed to prevent $35,000 in material handling injuries per year. Will the finance department approve your request, assuming an annual interest rate of 8%?
1) Yes
2) Not enough information
3) No
4) Only if the payback is extended to 36 months

Ans.1 This question requires calculating the present worth of the future savings. That is, what is the present value of $35,000 in loss that will be prevented each year over the next two years required by this finance department.

PW(1) = \$35,000 $(1 + 0.08)^{-1}$
= \$32,407
PW(2) = \$35,000 $(1 + 0.08)^{-2}$
= \$30,006

Total savings after 2 years = PW(1) + PW(2) = \$62,413
Since the present value of the savings is greater than the cost, your request should be approved.

585. The finance department requires that all investments pay for themselves in 24 months. You have recommended that \$50,000 in material handling equipment be installed to prevent \$35,000 in injuries per year. Will the finance department approve your request if the annual interest rate is 12%?

1) Yes
2) Not enough information
3) No
4) Yes if the payback period is extended to 48 months

Ans.1 This question requires calculating the present worth of the future savings (in this case, the accident costs of \$35,000 per year).

PW(1) = \$35,000 $(1 + 0.12)^{-1}$
= \$31,250
PW(2) = \$35,000 $(1 + 0.12)^{-2}$
= \$27,902

Total savings = PW(1) + PW(2) = \$59,151 which is STILL greater than \$50,000. Therefore, this request ought to be approved, too.

586. In conducting a safety and health audit you ask for workers' compensation cost data. You find the location's experience modification rate to be 0.55. How would you rate the location's safety and health performance with respect to this measure?

1) Excellent
2) Average
3) Slightly below average
4) Poor

Ans.1 The experience modification is developed from the location's injury/illness frequency and severity rate and the industry rate. If the plant had the same experience as the industry as a whole, the experience modification rate would be 1.0. Since this plant had an experience modification rate of 0.55, it would be considered excellent.

587. Workers' compensation insurance premiums are determined by three rating systems namely, retrospective rating, experience rating, and:

1) Modification factor
2) Manual (or state) rating
3) Loss exposure rating
4) Prospective experience rating

Ans.2 The first step in the rate making process is to determine a basis for potential loss. With workers' compensation, it is obtained from the payroll (i.e., the number of employees), the number of hours worked, and pay rate. Different types of work entail various degrees of hazard. For example, an office worker will be insured with less risk than a machine operator based on their job classifications. Similar industries have similar or matching classifications. Manual rates are governed by the experience of the industry and are reported annually by insurance carriers.

588. As the manager of the safety and health function, you have access to cost data relating to insurance, medical, wages, and operations. How would you determine the total cost of an injury or illness?

1) Sum the insured and non-insured costs
2) Combine insurance cost for medical and compensation paid to claimant
3) Determine the direct cost and multiply it by four
4) Determine the direct cost and add 25%

Ans.1 Accident costs include both those that are insured and those that are not insured. The insured costs include medical and compensation costs paid to the claimant by the insurance company. The non insured costs include time lost by others who observed or rendered assistance at the time of the accident, time lost to investigate the accident, time to train a replacement, cost for damaged materials, etc.

589. In one year, a company with 65 employees recorded in their Form OSHA No. 300: 7 injury cases with lost workdays, 2 injury cases without lost workdays, and 2 illness cases without lost work days. The total hours worked by all employees is 130,000. What is the incidence rate of recordable injuries for that year?

1) 6.2 cases/yr
2) 11.0 cases/yr
3) 13.9 cases/yr

4) 16.9 cases/yr

Ans.4 Computation of incidence rate for OSHA-recordable cases:
= (No. of recordable cases * 200,000) / Total employee hours
= (11 cases * 200,000 hr/yr) / 130,000
= 16.92 cases/yr

The OSHA incidence rate is based on 100 employees working 2,000 hours per year. This is the basis for the 200,000 hr/yr in the equation.

590. In one year, a company with 65 employees recorded in their Form OSHA No. 300: 7 lost workdays from injuries, 2 injury cases without lost workdays, and 2 illness cases without lost work days. The total hours worked by all employees was 130,000. What is the incidence rate of lost workday injuries for that year?
1) 6.2 cases/yr
2) 7.0 cases/yr
3) 10.0 cases/yr
4) 10.8 cases/yr

Ans.4 Computation of incidence rate for lost workday cases:
= (No. of lost workday cases x 200,000)/Total employee hours
= (7 cases x 200,000 hr/yr) / 130,000 hr
= 10.77 cases/yr

The OSHA incidence rate is based on 100 employees working 2,000 hours per year. This is the basis for the 200,000 hr/yr in the equation.

591. Determine the OSHA-recordable injuries incidence rate for a company if the firm sustained 40 OSHA-recordable injuries in a year in which the 750-person firm worked 1,500,000 hours.
1) 5.3 cases/yr
2) 8.7 cases/yr
3) 10.2 cases/yr
4) 11.5 cases/yr

Ans.1 Computation of incidence rate for OSHA-recordable cases:
= (40 cases * 200,000 hr/yr) / 1,500,000 hr
= 5.3 cases/yr

592. A worker can perform a certain task in 16 hours. The part of the task the worker can do in H hours is:
1) 16 / H

2) H / 16
3) 16 - H
4) 16H

Ans.2 In one hour, the worker can do 1/16 of the total task. In H hours, the worker can do H times (l/16) = H / 16 of the task.

593. Just-in-time (JIT) inventory control seeks to:
1) Minimize the amount of raw material inventory in the facility at any time
2) Provide line employees with exactly what they need when they need it
3) Minimize procurement anomalies by establishing standard schedules (based on need) for delivering raw materials/spare parts
4) All of the above

Ans.4 JIT inventory control also helps to reduce overhead costs by reducing storage/warehousing requirements. JIT requires a company to fully understand its production and maintenance system so that optimizations can be made while enhancing quality.

594. Zero-based budgeting means:
1) Eliminating the budget
2) Developing a budget without considering the value of last year's budget
3) Creating a budget with no basis
4) None of the above

Ans.2 The essence of zero-based budgeting is starting at "zero", determining which programs are critical, and assessing how much each program actually costs. Last year's budget is not relevant, except as a data point. True zero-based budgeting sometimes shows that some programs are under-funded, while others are not critical or are duplicative.

595. Crew Resource Management (CRM) is a technique to properly use all of the operations professionals on a team to reliably meet mission objectives. It is derived from efforts to make aircraft cockpit crews more team-oriented versus the highly authoritarian environments in which cockpit crews formally found themselves. Of the following, which dynamic human environment is the best candidate for applying the principles of CRM?
1) Offsite incident command staff-onsite incident command staff during an emergency
2) Trainer-trainees during a class
3) Employee-supervisor during a performance appraisal
4) Passengers-flight attendants during a flight

Ans.1 Incident command could greatly benefit from applying the principles of CRM. In this environment, all of the incident command professionals have the same goal; however, culture, work habits, and other non-relevant issues can cloud the efficient exchange of information to meet the mission objectives.

596. Theory X management assumes the following:
1) The worker is basically uninterested in work
2) Worker must be motivated to work
3) Work is natural and desirable
4) Both 1 and 2

Ans.4 Theory X management holds that workers must be made to work by a set of external rewards and punishments. Work is not a natural condition.

597. Theory Y management assumes the worker is:
1) Basically uninterested in work
2) Basically interested in work
3) Basically blase towards work
4) Basically accident prone

Ans.2 Theory Y management holds that the worker is interested and motivated to work, as opposed to Theory X.

598. According to contemporary motivation theory, the strongest motivator is:
1) Fear of a supervisor
2) Status
3) Recognition of achievement
4) Pay

Ans.3 Current theory holds that recognition of achievement is the single most important motivator.

599. The LEAST effective means for involving employees in process safety management at a facility is through:

1) Reward and discipline
2) Training
3) Safety and health committees
4) Review and input to the program

Ans.1 An individual reward and discipline program does less to encourage involvement than employee training, committees, and review/input to the program. The latter three allow employees to become informed and involved up-front in decisions that affect their safety and health.

600. Responsibility for safety at the operational level rests with:
1) General managers
2) Safety directors
3) Facilities personnel
4) Line supervisors

Ans.4 Line supervisors are responsible for safety. Safety management is among the other key aspects of a supervisor's job duties.

601. What functions do corporate safety groups normally NOT perform?
1) Safety policy development
2) Safety procedure development
3) Corporate safety manual
4) Investigate plant accidents

Ans.4 Corporate safety groups do not normally investigate plant accidents. They frequently review the findings, root causes, and corrective actions generated from a formally conducted incident investigation, though.

602. It can be difficult to get non-users of seat belts to wear them because:
1) The positive consequence of wearing them is future and uncertain
2) The negative consequences of wearing them are immediate and certain
3) Both 1 and 2
4) None of the above

Ans.3 Since the positive consequences of wearing seat belts (i.e., prevent injury) is future and uncertain, and the negative consequences of wearing them are immediate and certain (uncomfortable and it takes time), some people may be motivated not the wear them.

E. Scott Geller, The Psychology of Safety

603. In behavioral terms, safety management can be considered a continuous fight against human nature because:
   1) Positive consequences of unsafe behavior tend to be immediate and certain
   2) Negative consequences of unsafe behavior tend to be uncertain
   3) Both 1 and 2
   4) None of the above

Ans.3 The positive consequences of unsafe behavior tend to be immediate and certain, for example, more comfort in not wearing personal protective equipment. In addition, the negative consequences of unsafe behavior, such as having an accident, are uncertain.

E. Scott Geller, The Psychology of Safety

604. Why is it a good idea to have a formal safety and health statement of policy?
   1) Safety and health should be of primary concern to everybody
   2) It is the starting point for all activities in the safety and health program
   3) It expresses management's attitude toward safety and health
   4) It publicizes the safety and health function

Ans.3 The formal safety and health statement of policy is an endorsement of top management's commitment and support. It is management's expression of the direction to be followed.

605. A good company safety policy will include all of the following EXCEPT:
   1) Responsibility
   2) Accountability
   3) Authority
   4) Accident history

Ans.4 A company safety policy should include: responsibilities, accountabilities, authority, scope, intent, and standards. Though accident history is interesting, a company's safety policy

should not dwell on specific past events, but should indicate management's expectations toward future safety-related performance.

606. What is the principle focus of an in-plant medical program?
1) Diagnosis
2) Treatment
3) Prevention
4) Rehabilitation

Ans.3 The focus of medical programs is on the prevention of occupational disease.

607. The concept of "hierarchy of needs" was developed by:
1) Freud
2) Maslow
3) Jung
4) MacGregor

Ans.2 Maslow's hierarchy of needs is a critical component of many management theories dealing with human needs.

608. A supervisor's attitude toward safety and health _____ influence on those that he or she supervises.
1) Has little
2) Has significant
3) Has no
4) May have

Ans.2 A supervisor's attitude is of critical importance to an employee's perception of job safety and health.

609. There are several management theories of motivation, such as Theory X and Theory Y. Which one of the following is NOT true?
1) Theory X - The worker is uninterested and unmotivated to work
2) Theory Y - The worker is interested and motivated to work
3) Job Enrichment Theory of Herzberg - Work environment, benefits, and work conditions are all-important
4) Job Enrichment Theory of Herzberg - Work must be meaningful, must provide recognition, give responsibility, and allow for professional growth in order to motivate the worker

Ans.3 Job Enrichment Theory of Herzberg - Work environment, benefits, and work conditions are all-important incorrectly identifies Herzberg's Job Enrichment Theory.

Herzberg believed that hygiene issues (such as work environment, benefits and work conditions) are necessary before motivation and resultant productivity increases can occur but do not result in motivation. Instead, workers are motivated by work that is meaningful, provides recognition, gives responsibility, and allows for professional growth.

610. The process of determining safety goals, allowing supervisors a chance to perform, providing assistance along the way, and rewarding them according to their progress is known as:
    1) Management by Exception
    2) Safety by Objectives
    3) OSHA Star Program
    4) Motivation-Hygiene Theory

Ans.2 Safety by Objectives is the process of determining safety goals, allowing supervisors a chance to perform, providing assistance along the way, and rewarding them according to their progress.

611. The key to effective line safety performances are management procedures that fix:
    1) Accountability
    2) Flexibility
    3) Optimization
    4) Variability

Ans.1 To be effective, line managers must be accountable for safety performance. Accountability is critical.

612. In the study of traditional management, the term "Span of Control" is used to indicate:
    1) The control of many by one
    2) The number of personnel under one supervisor
    3) The number of supervisors that should be in control
    4) The number of supervisors actually in control

Ans.2 Span of control is the number of personnel that directly report to a single supervisor.

613. The best way to create acceptable new responses to environmental stimuli is to:

1) Discipline when not performed properly
2) Provide training
3) Reward the desired response
4) Write a mandatory procedure

Ans.3 Rewarding a desired response is the best way to encourage the response in the future.

Dan Petersen, Safety Management, A Human Approach

614. Which of the following is NOT generally considered positive reinforcement?
1) Money
2) Praise
3) Recognition
4) Training

Ans.4 Money, praise, and recognition are forms of positive reinforcement.
Dan Petersen, Safety Management: A Human Approach

615. Total quality management (TQM) is an overall commitment to continuously improve quality performance at all levels. All of the following concepts apply EXCEPT:
1) Empowerment
2) Improvement
3) Central thinking
4) Auditing

Ans.3 Central thinking is not a concept of TQM, although acting as one voice is a concept of TQM.

616. Communication means conveying a message and having it understood. Retention of the safety message is best retained when it is:
1) Presented orally at a rate of 120 words/min
2) Presented orally at a rate of 450 words/min with body movement
3) Presented orally at a rate of 120 words/min with visual aids
4) Presented orally at a rate of 120 words/min with hand and face gestures

Ans.3 We learn 87% through our eyes and only 7% through our ears. (The remaining 6% balance occurs from other senses, e.g. touch and smell.) Also, there are differences in the rates of listening versus talking and understanding. The average speech rate is 120

words/min, while the rate of understanding is 450 words/min. Adding visual aids to an oral presentation increases the rate of learning and retention of the message presented.

617. A management technique frequently used to explore options and to determine the most remote contingencies is:
   1) Brainstorming
   2) “One on One”
   3) Staff meeting
   4) Retreat meeting

Ans.1 Brainstorming is normally used to fully develop a situation's potential impact. It normally involves several people in a setting designed to draw out all ideas. The ideas are then sorted out and prioritized, and action plans are developed. The unstructured nature of this activity tends to produce more creative ideas. A trained facilitator is often used.

618. The goal of ergonomic/human factors engineering (HFE) may be any of the following EXCEPT:
   1) Increase worker's compensation costs
   2) Reduce worker errors and accidents
   3) Control stresses affecting worker's psychic comfort
   4) Increase worker productivity

Ans.1 Ergonomics and human factors engineering in the United States is a multidisciplinary activity. It deals with interactions between individuals and their total working environment, plus stresses relating to such environmental elements as atmosphere, heat, light, and sound, in addition to workplace tools and equipment.

619. One of the current principles of safety management states that the key to effective line safety performance is having management procedures that:
   1) Fix accountability for accidents
   2) Identify and correct unsafe acts and or conditions
   3) Improve worker attitude through education and training
   4) Engineer the hazard out of operations

Ans.1 Fixing accountability is based on the belief that a line manager will achieve results only in those areas in which he or she is being measured. It has long been felt by professional safety practitioners that the lack of procedures for fixing accountability is safety's greatest failing.

620. One of the principles of safety management states that certain sets of circumstances can be predicted to produce severe injuries. Which of the following circumstances does NOT apply?

1) High energy sources
2) Nonproductive activities
3) Unusual, non-routine activities
4) Accidents occur by chance and cannot be predicted

Ans.4 The old approach to safety states that once an accident occurs, the severity of injury is dictated by chance. Modern thinking asserts that certain sets of circumstances can be predicted to produce severe injuries. Essentially, we can attack severity directly instead of hoping the severe accident frequency is low.

621. In what ways should safety be managed differently from other company functions?

1) Cannot set goals
2) Personal contacts are more important
3) Budget support is not available
4) Managed the same

Ans.4 The management of a safety program should not differ from any other program - set goals, plan, organize, and execute the plan to achieve the goals. The more the safety program's goals and management style are understood, the more support there will be for the program.

622. What is the key to effective safety management?

1) Compliance with OSHA standards
2) Establishing accountability
3) Engineering controls of workplace hazards
4) Safety committees

Ans.2 Management must hold supervisors accountable for the unsafe acts committed by the employees they supervise. Accountability produces interest and attention. The result is a more effective program.

623. What is the most important factor in accident prevention?

1) Comprehensive, written accident prevention programs
2) Adequate training for all employees
3) Competent staff safety professionals
4) Dedicated management support

Ans.4 The most critical element of any accident prevention program is management support. Without this support the program cannot succeed.

624. Which of the following is the first step in establishing an effective process safety management program?
   1) Training employees and contractors
   2) Performing comprehensive hazard analyses
   3) Writing a process safety management program
   4) Establishing a company culture centered on safety

Ans.4 An effective process safety management program cannot be achieved without first establishing a culture in which safety is a priority and is part of the daily routine of the employee. The hazard analysis and written program will not be implemented, and employee training is not effective without the commitment of the work-force or management.

625. What is the primary reason for a company to institute a fleet safety program?
   1) To control insurance costs
   2) To prevent accidents
   3) To prevent liability claims
   4) To provide guidance to employees

Ans.2 The primary reason for a company to institute a fleet safety program is to prevent accidents. As a result of preventing accidents, the number of liability claims will decline and insurance costs should decrease.

626. There are similarities and differences between motor-vehicle safety and industrial safety. Which similarity stands out?
   1) Supervisory control
   2) Environment control
   3) Management's policy
   4) Employee policy

Ans.3A statement of management's intent bears the same importance in motor-vehicle accident control as it does in industrial safety. Generally, the policy spells out: Management's statement that safety on the road is important; the company safety program includes the driver; the driver will support the program: and specific safety responsibilities will be assigned to management. Motor-vehicle safety is different from industrial safety in two ways: On-the-road employees are not under constant supervision as they would be in a plant, nor can the driving environment be controlled as effectively as it can be within an industrial setting.

627. A generic process of evaluating competing courses of action by examining the dollar costs of certain abatement actions versus the dollar value of the benefits received is called:
1) Prime cost assessment
2) Cost analysis
3) Cost-benefit analysis
4) Risk management

Ans.3 Cost-benefit analyses are generic processes for evaluating competing courses of action by examining the dollar costs of certain actions versus the dollar value of the benefits received.

628. A measure of the projected dollar loss per unit (or prorated per unit) calculated by dividing the total risk exposure by the total number of units produced is called:
1) Unit loss
2) Unit loss exposure
3) Unit risk exposure
4) Risk unit

Ans.3 Unit risk exposure is a measure of the projected dollar loss per unit (or prorated per unit) and is calculated by dividing the total risk exposure by the total number of units produced.

629. The ratio of the risk exposure to the project cost is called:
1) Loss ratio
2) Cost ratio
3) Risk exposure ratio
4) Risk ratio

Ans.3 Risk exposure ratio is the ratio of the risk exposure to the project cost.

630. If a company is using self insurance to handle risk, the company:
1) Is using the insurance of the employees to cover their risk
2) Is not insured
3) Is buying the insurance from a state plan
4) Is carrying no deductible

Ans.2 The company is not insured and must "fund" their own insurance plan. Self insurance can be used for various types of risks, including worker's compensation, but often the state

will require the company to be in good financial health to use this approach, or be bonded for the exposure.

631. Your company manufactures and fills small propane cylinders that are used for torches, camp stoves, and gas-powered lanterns. This product is packaged and distributed to wholesalers and retailers using a combination of transportation options including the company's own fleet of trucks. Your company is responsible for the:
   1) Safe containerization of the product shipped via the company fleet. For other modes of shipment, safe containerization is the responsibility of the receiver of the cylinders
   2) Development of safe containers and transportation specifications for the product
   3) Safety of manufacturing, handling, and storage of the product within the facility, and the safe containerization for shipment
   4) Safety of manufacturing. The shipper is responsible for containerization

Ans.3 The company is responsible for safety in manufacturing and for safe containerization for shipping no matter who does the shipping.

632. As an effective manager, how would you discipline an employee's first minor safety rule infraction?
   1) Oral reprimand
   2) Written reprimand
   3) Ignore it
   4) Suspension

Ans.1 The most logical way to deal with a minor rule infraction is some sort of oral reprimand for the first offense. Ignoring it will not cause a change in employee behavior, and a formal reprimand/suspension will likely cause employee dissention.

633. How can a Safety Director LEAST influence line accountability for safety?
   1) Charge accidents to departments
   2) Put safety in the supervisor's appraisal
   3) Have safety affect the supervisor's income
   4) Put the supervisor on warning

Ans.4 All of the above actions will make line supervisors accountable for the safety of their group. However, "putting a supervisor on warning" will cause an adversarial relationship which benefits no one. Once accountability is assigned, improvement in performance is

attained more readily. This is true in any line function. Human nature is to pay attention to those things that will be measured.

634. What is the reason that line employees fail to report near misses to supervisors and/or safety staff?
1) Fear of job reprisals from reporting
2) Fear of peer pressure from reporting
3) Perception that nothing ever happens to near miss reports
4) All of the above

Ans.4 There are several other reasons why employees do not report near misses. A critical component to an effective loss control program is removing these barriers to reporting near misses. Without effective near miss reporting, a loss control program will have difficulty evolving beyond reactive assessment.

635. In biosafety nomenclature, which agent Class is the most hazardous?
1) Class 1
2) Class 2
3) Class 3
4) Class 4

Ans.4 Biosafety agents range from Class 1 (safest) to Class 4 (most hazardous). Class 5 agents are foreign animal pathogens regulated by the USDA.

636. The color to indicate fire or danger is:
1) Red
2) Magenta
3) Green
4) Orange

Ans.1 Red is the color used to denote fire or danger. 29 CFR 1910.144 (a)(1)

637. The basic color used for designating caution is:
1) Red
2) Orange
3) Yellow
4) Blue

Ans.3 Yellow is the basic color for designating caution. Applications include: aisle marking, "caution" signs, construction tractors and bulldozers, handrails, guardrails, marking for low beams, pipes, and projections. ANSI Z53.1 (1967)
29 CFR 1910.144 (a)(3)

638. The color used to identify physical tripping hazards is:
1) Red
2) Blue
3) Green
4) Black/yellow

Ans.4 Black/yellow is normally used to identify physical tripping hazards. 29 CFR 1910.144 (a)(3)

639. The color generally utilized to denote information is:
1) Blue
2) Red
3) Yellow
4) Orange

Ans.1 Blue is the color utilized to denote information.

640. What color is generally used to identify moving machine parts?
1) Red
2) Magenta
3) Green
4) Orange

Ans.4 Orange is used to identify moving machine parts.

641. What color is generally used to identify safety facts or first aid?
1) Red
2) Magenta
3) Green
4) Orange

Ans.3 Green is the universal symbol for safety and first aid.

642. NFPA 704 provides a diamond shaped diagram shown for inherent hazards of a chemical. What does the color blue indicate?
1) Reactivity hazard
2) Flammability hazard
3) Combustibility hazard
4) Health hazard

Ans.4 The color blue stands for health hazard.

643. NFPA 704 provides a diamond shaped diagram shown for inherent hazards of a chemical. What does the color yellow indicate?
1) Reactivity hazard
2) Flammability hazard
3) Combustibility hazard
4) Health hazard

Ans.1 The color yellow relates to a reactivity hazard.

644. NFPA 704 provides a diamond shaped diagram shown for inherent hazards of a chemical. What does the color red indicate?
1) Reactivity hazard
2) Flammability hazard
3) Combustibility hazard
4) Health hazard

Ans.2 The color red indicates a flammability hazard.

645. ANSI A-13.1 provides a scheme for the identification of piping systems. What is the standard color used to identify protective materials?
1) Blue-yellow
2) Bright blue
3) Green
4) Red

Ans.2 Bright blue is the standard color used to identify protective materials.

646. ANSI Z-35.1 provides the specifications for industrial accident prevention signs. What is appropriate to indicate caution against lesser hazards?
1) Yellow background color; black lettering
2) Black lettering on white background
3) Light blue background color, black lettering
4) Black lettering on orange background

Ans.1 Yellow background color with black lettering is appropriate to indicate caution against lesser hazards.

647. ANSI A-13.1 provides a scheme for the identification of piping systems. What standard color is used to identify dangerous materials?
1) Red
2) Blue-green
3) Yellow
4) Orange

Ans.3 Yellow is the standard color used to identify dangerous materials according to ANSI A-13.1.

648. Rank the order of traditional hazard warning terms from most serious to least serious.
1) Caution, warning, danger
2) Danger, caution, warning
3) Danger, warning, caution
4) Warning, danger, caution

Ans.3 Traditionally, there are three levels of hazards: Danger - used where there is an immediate hazard, which, if encountered, will result in severe personal injury or death. Warning - used for hazards or unsafe acts which could result in severe personal injury or death. Caution - used for hazards or unsafe acts which could usually result in minor personal injury, product damage, or property damage.

649. Flammable storage cabinets are normally:

1) Blue
2) Magenta
3) Yellow
4) Green

Ans.3 Most flammable storage cabinets are painted yellow. In addition to being painted yellow they are labeled as Flammable Liquid Storage Cabinets. Some are also red.

650. A fire hydrant painted red on the bonnet and the caps usually indicates that the water flow rate from the hydrant is:

1) Less than 500 gpm
2) 500 to 999 gpm
3) 1,000 to 1,499 gpm
4) 1,500 gpm or greater

Ans.1 Red corresponds to less than 500 gpm.
Orange corresponds to 500 to 999 gpm.
Green corresponds to 1,000 to 1,499 gpm.
Light blue corresponds to 1,500 or greater gpm.
NFPA 291 - Recommended Practice for Fire Flow Testing and Marking of Hydrants

651. When walking through an office occupancy where the acoustic ceiling was 10 ft, you noticed that the automatic sprinkler heads hung down and were color coded white. What should this tell you?

1) It was an ordinary pendent sprinkler head with an operating temperature of 135ºF - 170ºF
2) It was an ordinary upright sprinkler head with an operating temperature of 175ºF - 225ºF
3) It was an intermediate pendent sprinkler head with an operating temperature of 175ºF - 225ºF
4) It was an intermediate pendent sprinkler head with an operating temperature of 135ºF - 170ºF

Ans.3 Sprinkler heads that hang down are called "pendent" and when color coded white signify an intermediate rating with a 175ºF - 225ºF operating temperature. NFPA 13 2-2.3.2 (Standard for the Installation of Sprinkler Systems)

652. You are conducting a formal safety inspection of a manufacturing plant with your newly hired safety engineer. During your inspection you observe an employee, without eye protection, working at a bench installing parts. This is not a hazardous operation but it is a posted "Eye Protection" area. Which one of the following is the best course of action?

1) Contact the supervisor and discuss the situation
2) Test the skills of your new safety professional by letting him or her handle the situation
3) Confront the employee and determine "Why" eye protection is not being used
4) Note the discrepancy and do not discuss it until the outbrief when the CEO and the supervisor are both present

Ans.1 Contact the supervisor who has control of the workplace and discuss the infraction. Communicate this process to the new safety professional.

653. While investigating a serious accident at your facility, you are approached by a journalist with a microphone. You should:

1) Politely ignore the journalist
2) Indicate to the journalist that he or she is interfering with an investigation
3) Stop and talk to the journalist
4) Direct the journalist to your supervisor

Ans.3 Journalists want the facts, and if you do not give them factual information, they may theorize and then collect evidence to verify these theories (regardless of the actual facts). In most cases, stopping and spending a minute or two with a journalist and providing him or her with appropriate facts is worthwhile for public goodwill. Passing off a journalist to your supervisor (who probably has fewer facts than you) will not be helpful.

For most major accidents, the organization sets up a dedicated public affairs area for journalists. If this is the case, you may want to direct the inquiring journalist there; however, stopping and talking with the lone journalist is still a pretty good tactic.

654. You are the safety manager in a large manufacturing facility. According to company policy, the safety office must approve any activity that is not classified as a routine operation. This has led to the safety office directly approving activities such as truck loading and unloading activities, machine maintenance activities, and custodial activities. Your chief compliant with this policy is that it:

1) Fails to make line managers accountable for their own operations
2) Requires the safety office staff to be larger than is necessary for a facility of this size
3) Requires a huge volume of record keeping and auditing
4) Prevents the safety office staff from focusing on more important activities

Ans.1 If the "safety office" is the final authority for approving these activities, then line supervisors have little (if any) stake in the work. The principles of accountability cannot be maintained in this environment.

655. The analytical technique called 5-Whys is used to identify:

1) Root causes of incidents
2) Human behavior characteristics
3) Behavior-based safety rules
4) Human errors

Ans.1 5-Whys is a qualitative analytical technique to identify the root causes of an incident sequence. Once the causal factors (discreet human errors and equipment malfunctions that led to the loss event) are known, a 5-Whys analysis can be performed on each causal factor to determine "Why" the causal factor was allowed to exist.